CS-503
Visual Intelligence

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Lecture 2

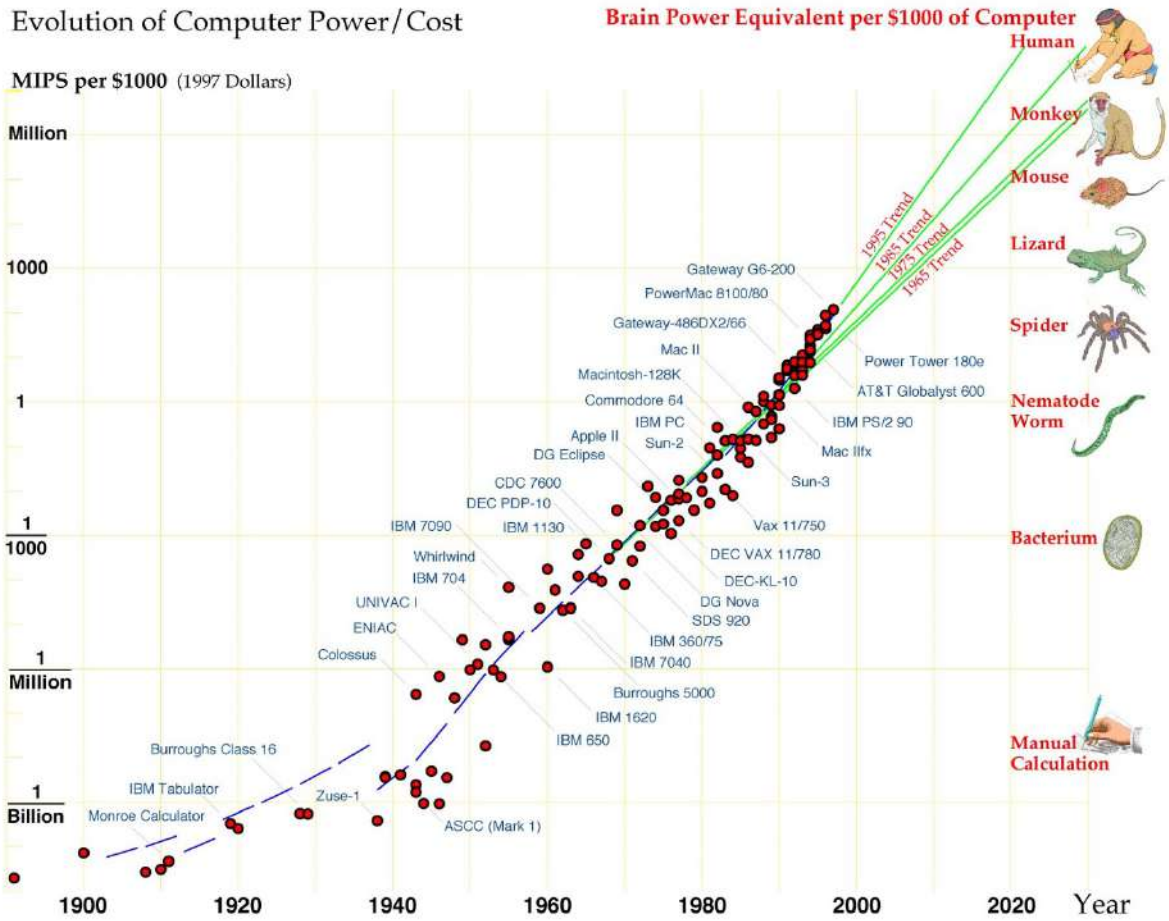
École
polytechnique
fédérale
de Lausanne

Recaps relevant to vision today

- **Data-driven/learning:**
  - e.g., Neural networks, AlexNet, etc.
- **Non-data-driven**
  - e.g., Image formation model, Image transformation, etc.
- “Method” recap, as opposed to historical credit assignment.

Fast Historical Recap

Bigger picture



Moravec 1998

Historical review (neural networks)

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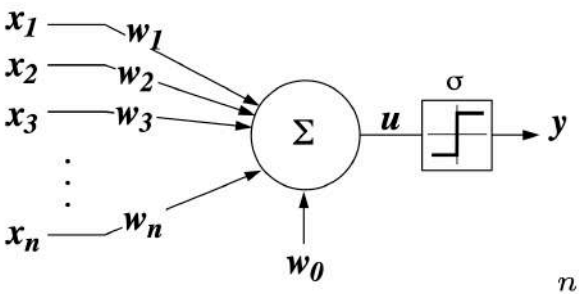
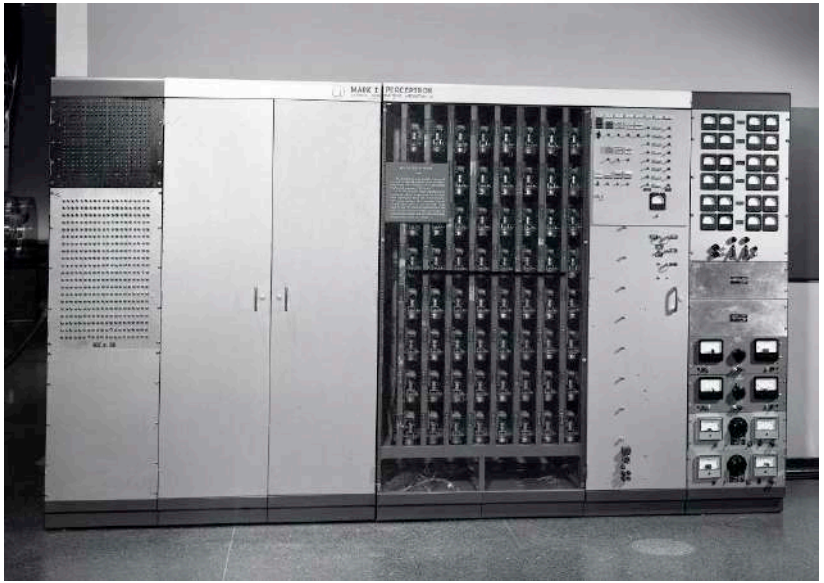
- Perceptron model, Rosenblatt, 1958.



$$u = w_0 + \sum_{i=1}^{n} w_i \, x_i$$

$$y = \sigma(u)$$

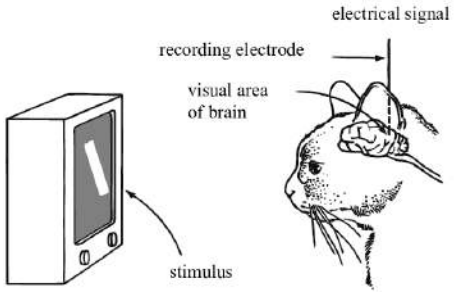


Historical review (neural networks)

- Perceptron model, Rosenblatt, 1958.

Historical review (neural networks)

- Hubel and Wiesel, ~1962.



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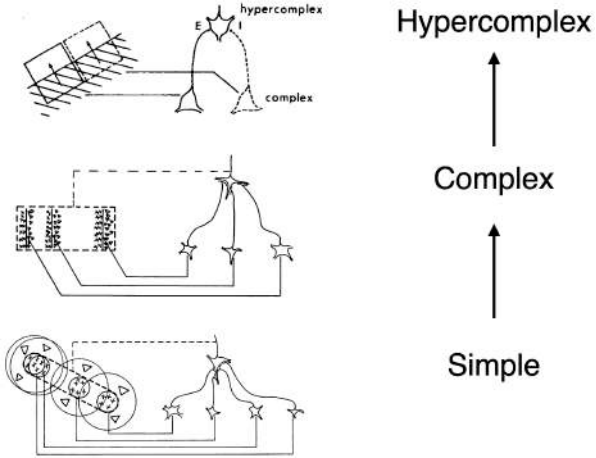

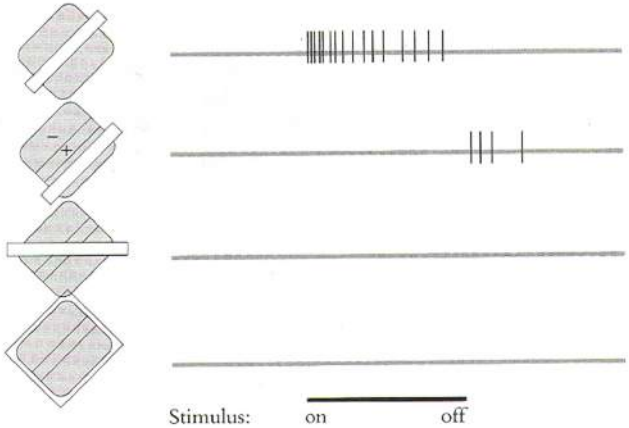

Historical review (neural networks)

- Neocognitron, Fukushima, 1980.
  - Modeled after Hubel and Wiesel.
  - Convolutional.
  - Multi-layer.
  - Hebbian Learning (no backpropagation).

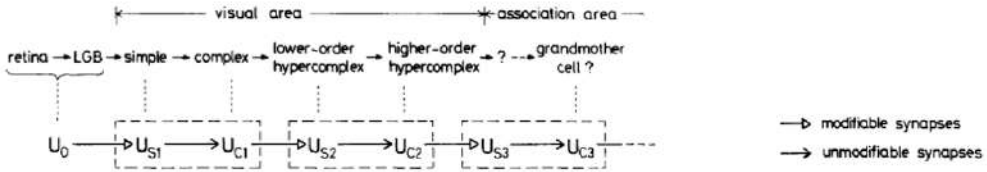

Fig. 1. Correspondence between the hierarchy model by Hubel and Wiesel, and the neural network of the neocognitron

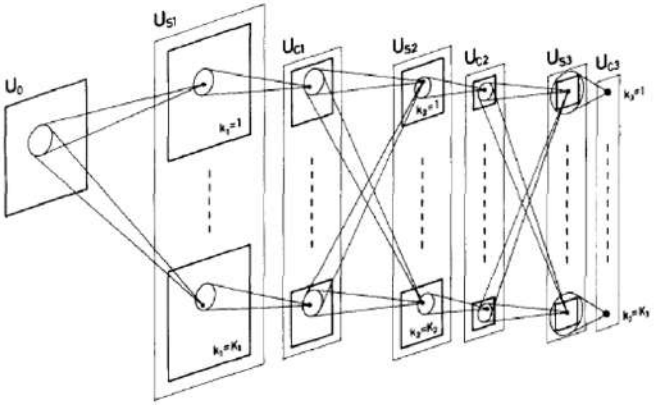

Fig. 2. Schematic diagram illustrating the interconnections between layers in the neocognitron

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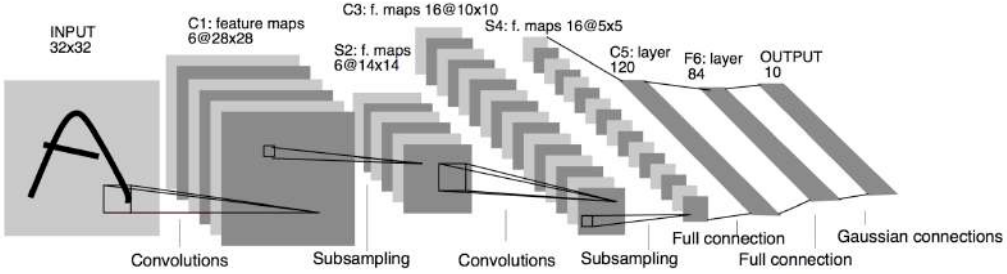
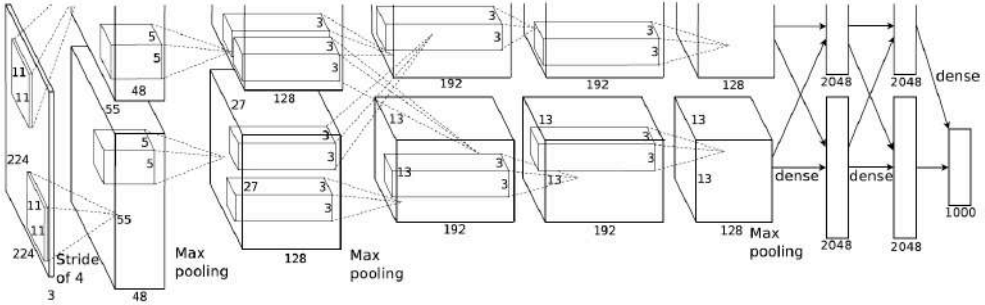
Historical review (neural networks)

- Backpropagation based learning. LeCun, Hinton, and co ~1989.
  - Backpropagation, as a method, already existed.
  - Most successful for handwritten digit recognition. Didn't work for standard vision tasks.



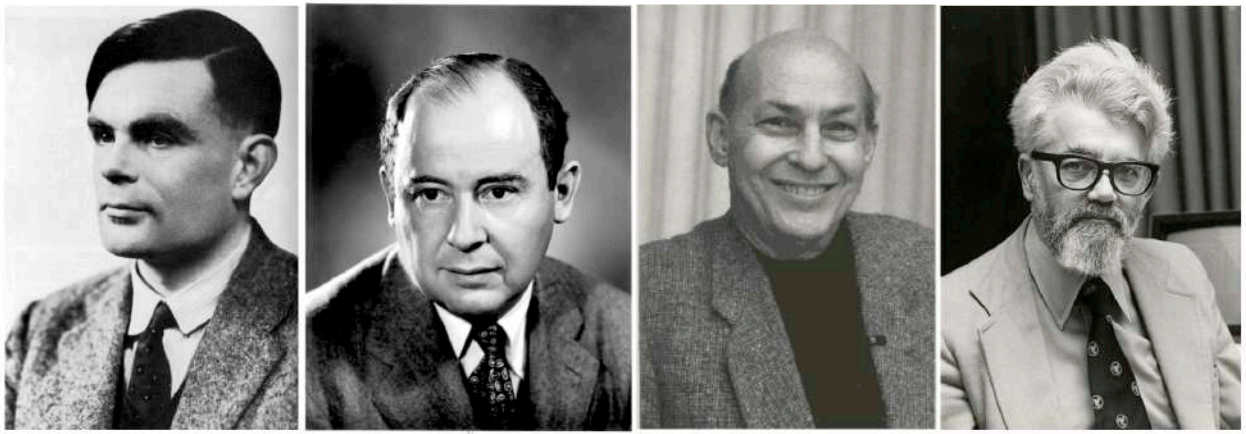
Historical review (neural networks)

- AlexNet, Krizhevsky, Sutskever & Hinton. 2012
  - Success at a standard vision task (ImageNet)
  - Deep Learning wave.
  - GPUs



Historical review

- "AI"



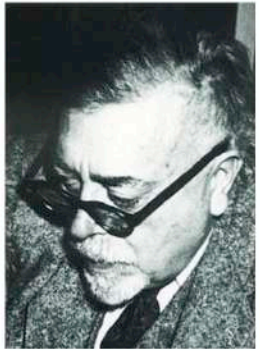
Alan Turing | John von Neumann | Marvin Minsky | John McCarthy

Among the most challenging scientific questions of our time are the corresponding analytic and synthetic problems: How does the brain function? Can we design a machine which will simulate a brain?
-- *Automata Studies*, 1956

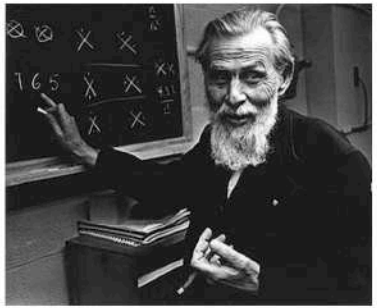
B. Olshausen

Historical review

- "Cybernetics"



Norbert Wiener



Warren McCulloch & Walter Pitts

Frank Rosenblatt

"The theory reported here clearly demonstrates the feasibility and fruitfulness of a quantitative statistical approach to the organization of cognitive systems. By the study of systems such as the perceptron, it is hoped that those fundamental laws of organization which are common to all information handling systems, machines and men included, may eventually be understood." -- Frank Rosenblatt

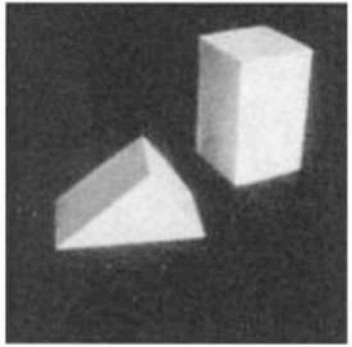
The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain. In, *Psychological Review*, Vol. 65, No. 6, pp. 386-408, November, 1958.

B. Olshausen

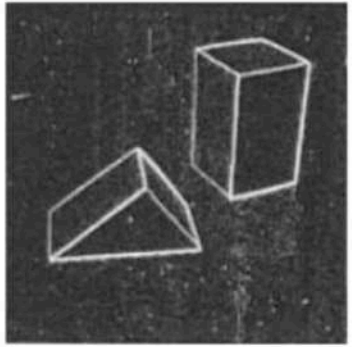
Historical review (computer vision)

- Larry Roberts Thesis 1963.
  - “Machine Perception of Three-Dimensional Solids”

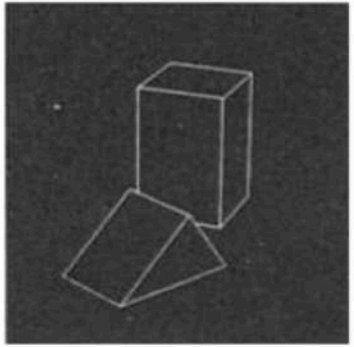
Larry Roberts
“Father of Computer Vision”



Input image



2x2 gradient operator



computed 3D model
rendered from new viewpoint

Historical review (computer vision)

- Larry Roberts Thesis 1963.
- The Summer Vision Project 1966

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

PROJECT MAC

Artificial Intelligence Group July 7, 1966
Vision Memo. No. 100.

THE SUMMER VISION PROJECT

Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

Historical review (computer vision)

- **1960s**: Birth
  - Larry Roberts Thesis 1963.
  - The Summer Vision Project 1966
- **1970s**: Foundational work on image formation
- **1980s**: Applied mathematics: geometry, multi-scale analyses
- **1990s**: Geometric analysis. Vision+graphics. Resurfacing of statistical learning.
- **2000s**: Progress in visual recognition. Pre-deep learning. DPM. PASCAL.
- **2010s**: Deep Learning.

J. Malik

Historical review AI

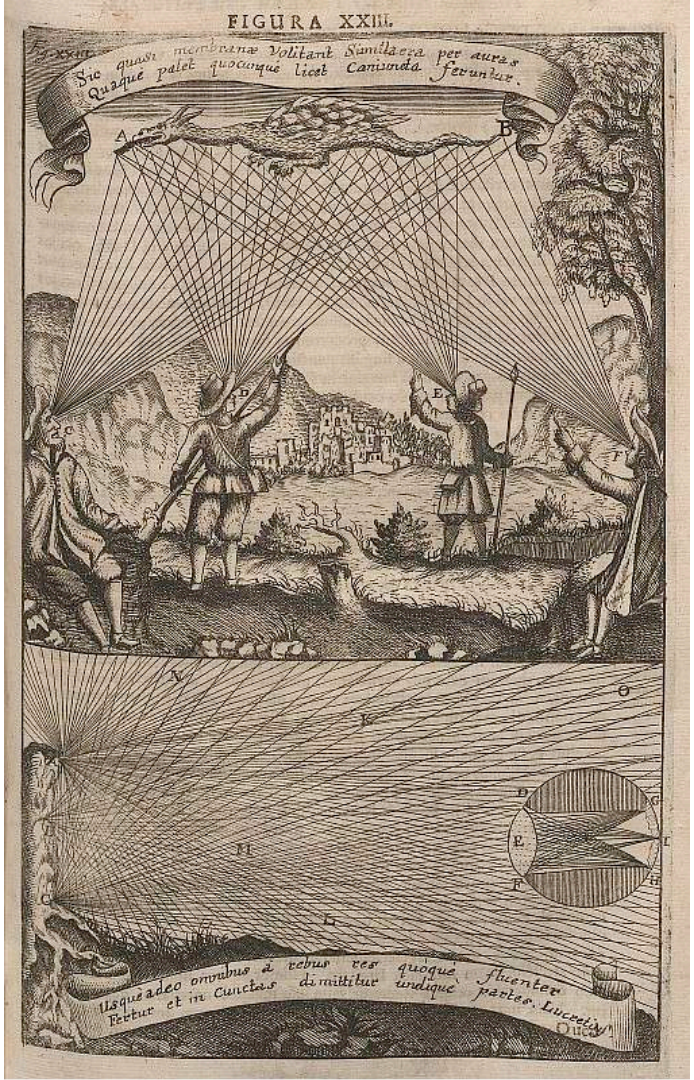
A good recap:

https://youtu.be/R3YFxF0n8n8

Fast Computer Vision Recap

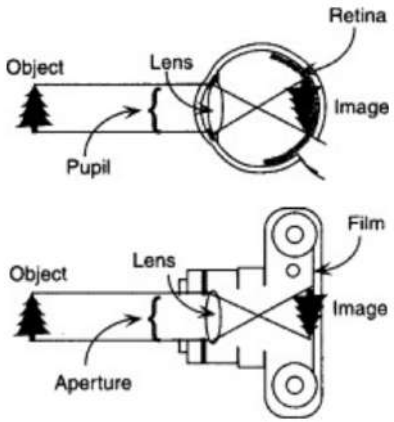
Vision

- Emission Theory (ca. 400 BC)



Vision

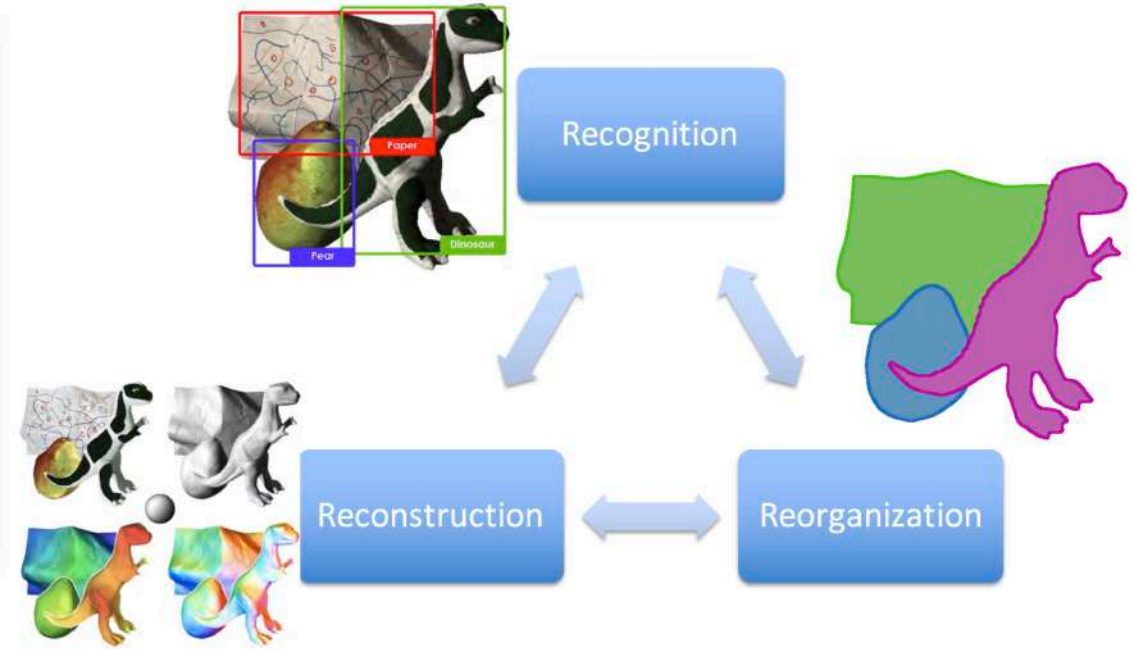
- defined as the process of acquiring knowledge about environmental objects and events by extracting information from the light they emit or reflect. [S. Palmer]
  - optical process: image formation
  - cognitive process: phenomena of visual perception
  - physiology of the visual nervous system: biological implementation



S. Palmer

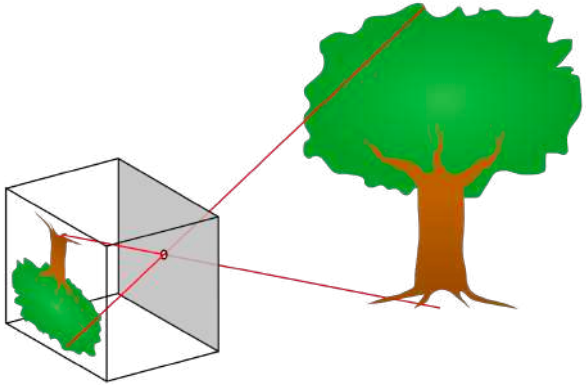
One useful organization

- Three Rs of Computer Vision.
- Interaction between the Rs.

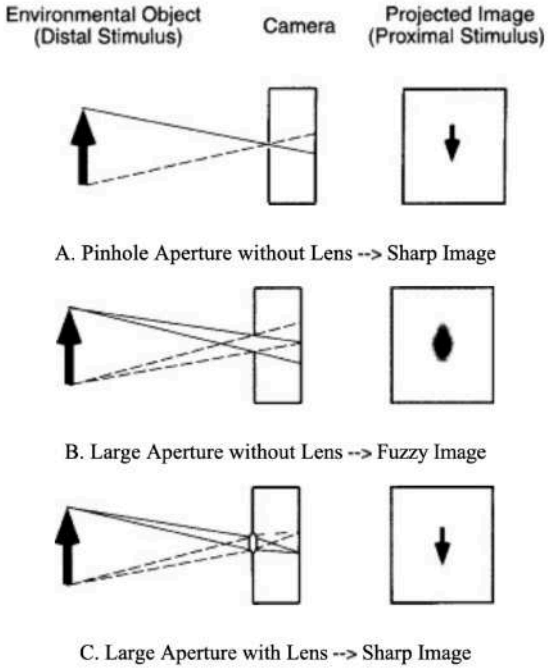


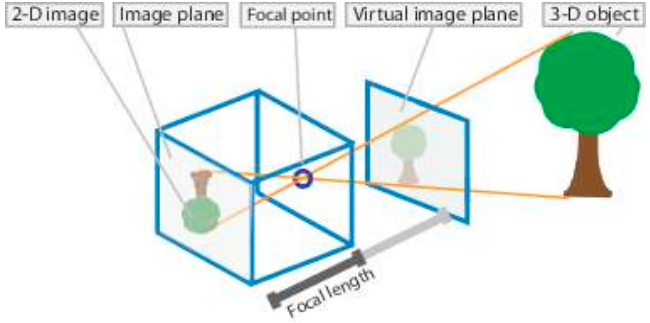
The three R's of computer vision: Recognition, reconstruction and reorganization. J. Malik et al. 2016.

Rudimentary Image Formation



Rudimentary Image Formation





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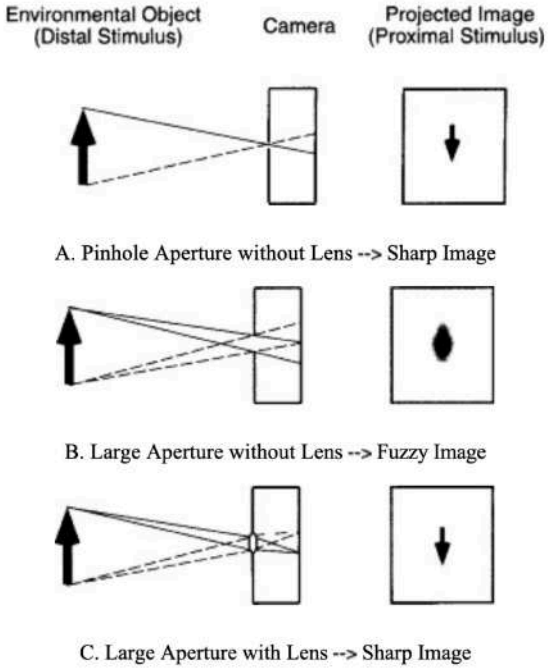

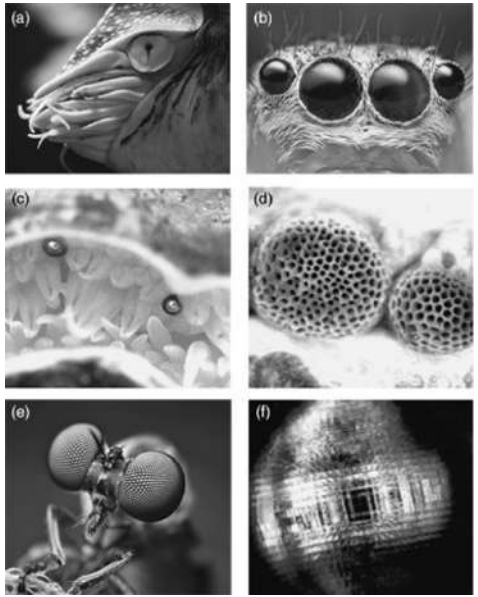

(a) The large pinhole eye of the cephalopod mollusc Nautilus. (b) Corneal eyes of a jumping spider Platycryptus. (c) Concave mirror eyes of the scallop Pecten. (d) Primitive compound eye of the ark clam Barbatia. (e) Compound eye of a male robberfly Holocephala. (f) Mirror compound eye of the shrimp Palaemonetes.

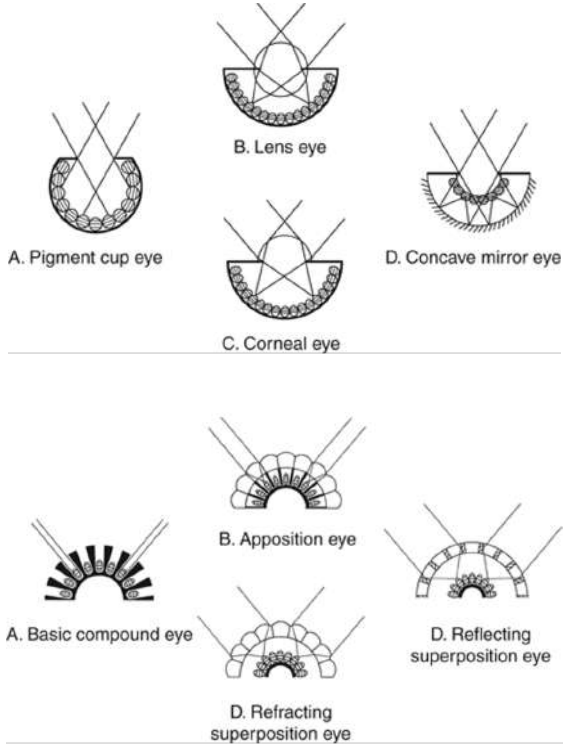

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Image Transformation

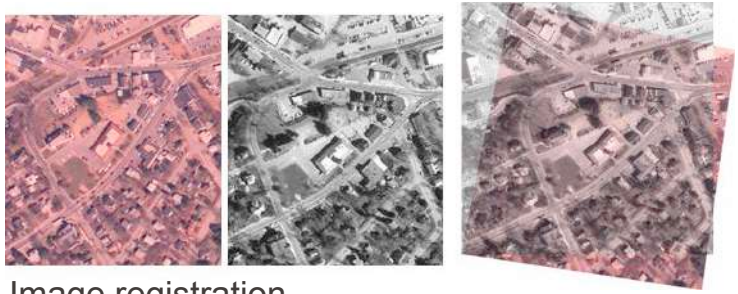


Image registration.

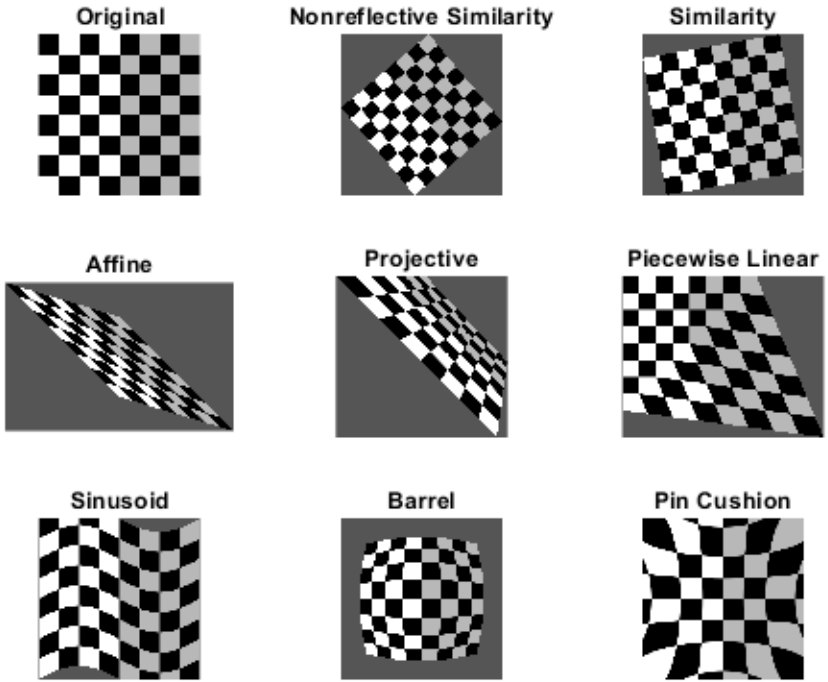
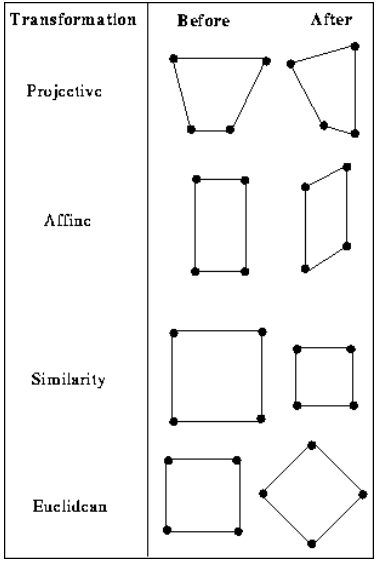
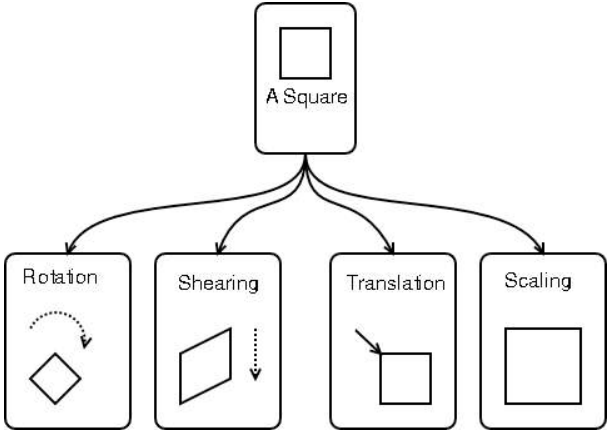


Image Transformation

Transformation	Matrix	# DoF	Preserves	Icon
translation	$\begin{bmatrix} I \mid t \end{bmatrix}_{2\times 3}$	2	orientation	
rigid (Euclidean)	$\begin{bmatrix} R \mid t \end{bmatrix}_{2\times 3}$	3	lengths	
similarity	$\begin{bmatrix} sR \mid t \end{bmatrix}_{2\times 3}$	4	angles	
affine	$\begin{bmatrix} A \end{bmatrix}_{2\times 3}$	6	parallelism	
projective	$\begin{bmatrix} \tilde{H} \end{bmatrix}_{3\times 3}$	8	straight lines	





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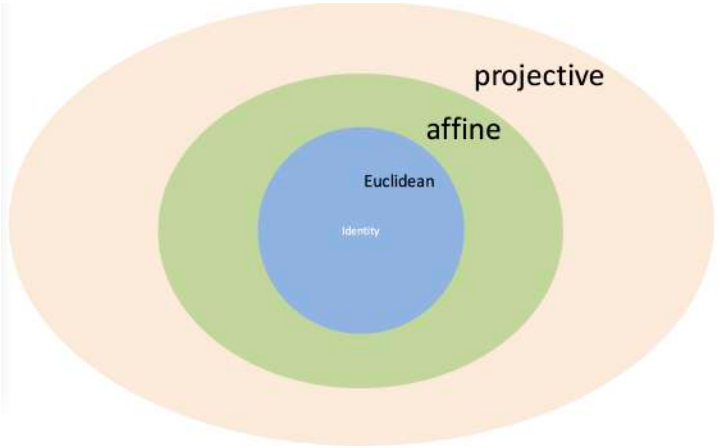
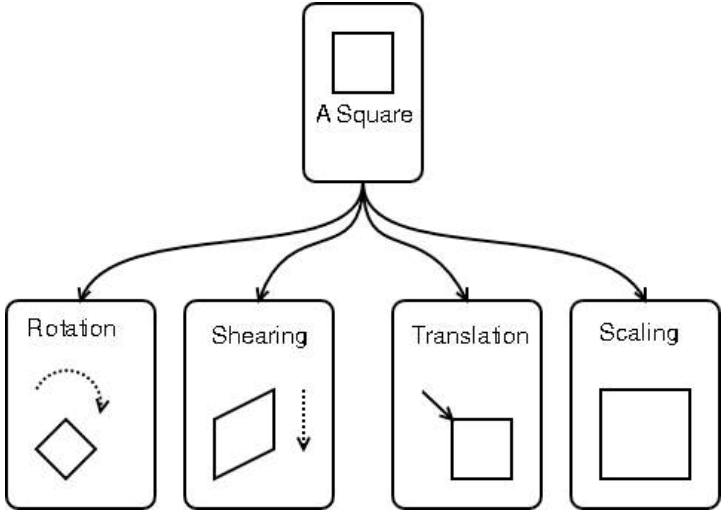




Correspondences

Correspondences

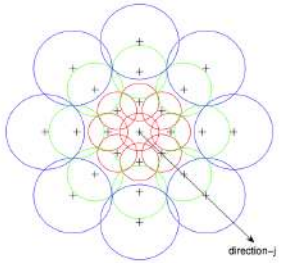
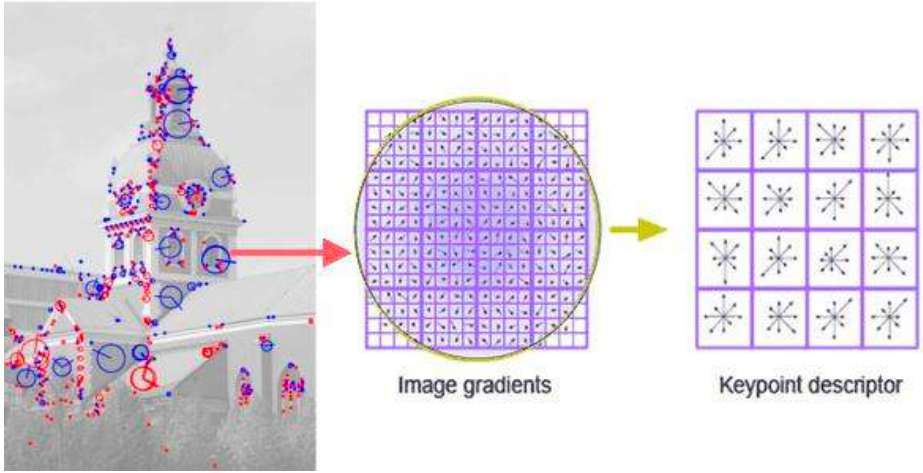


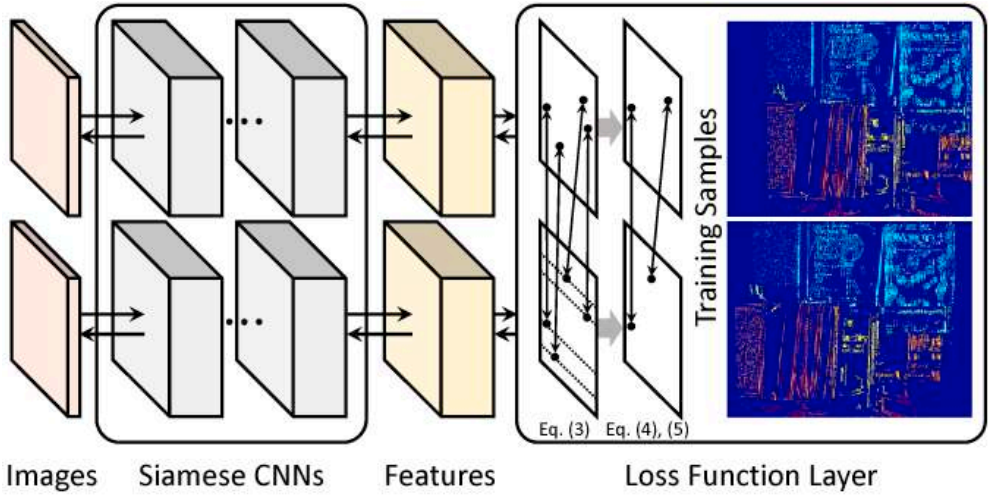
Figure 2. The DAISY descriptor. Each circle represents a region where the radius is proportional to the standard deviations of the Gaussian kernels and the '+' sign represents the locations where we sample the convolved orientation maps center being a pixel location where we compute the descriptor. By overlapping the regions we achieve smooth transitions between the regions and a degree of rotational robustness. The radii of the outer regions are increased to have an equal sampling of the rotational axis which is necessary for robustness against rotation.



SIFT, Lowe et al. 2003
DAISY, Tola et al. 2008

Correspondences

- Now gone deep.



e.g. Zagoruyko, S., Komodakis, N.: Learning to compare image patches via convolutional neural networks (2015).

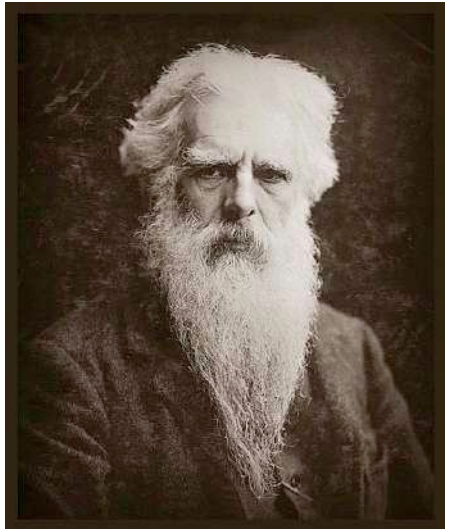
Dynamic Perspective

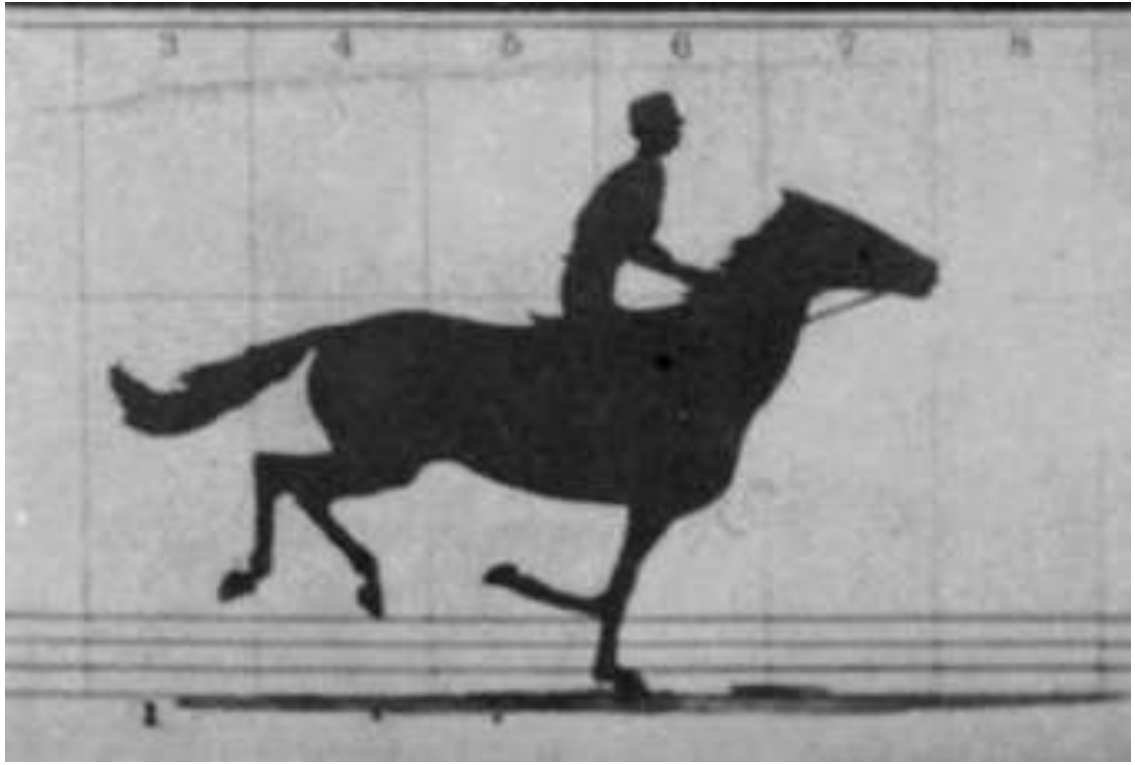
- Motion. Video.

Dynamic Perspective

- Motion. Video.

Eadweard Muybridge

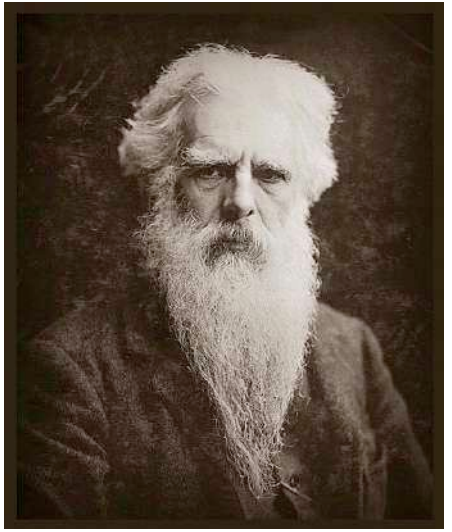


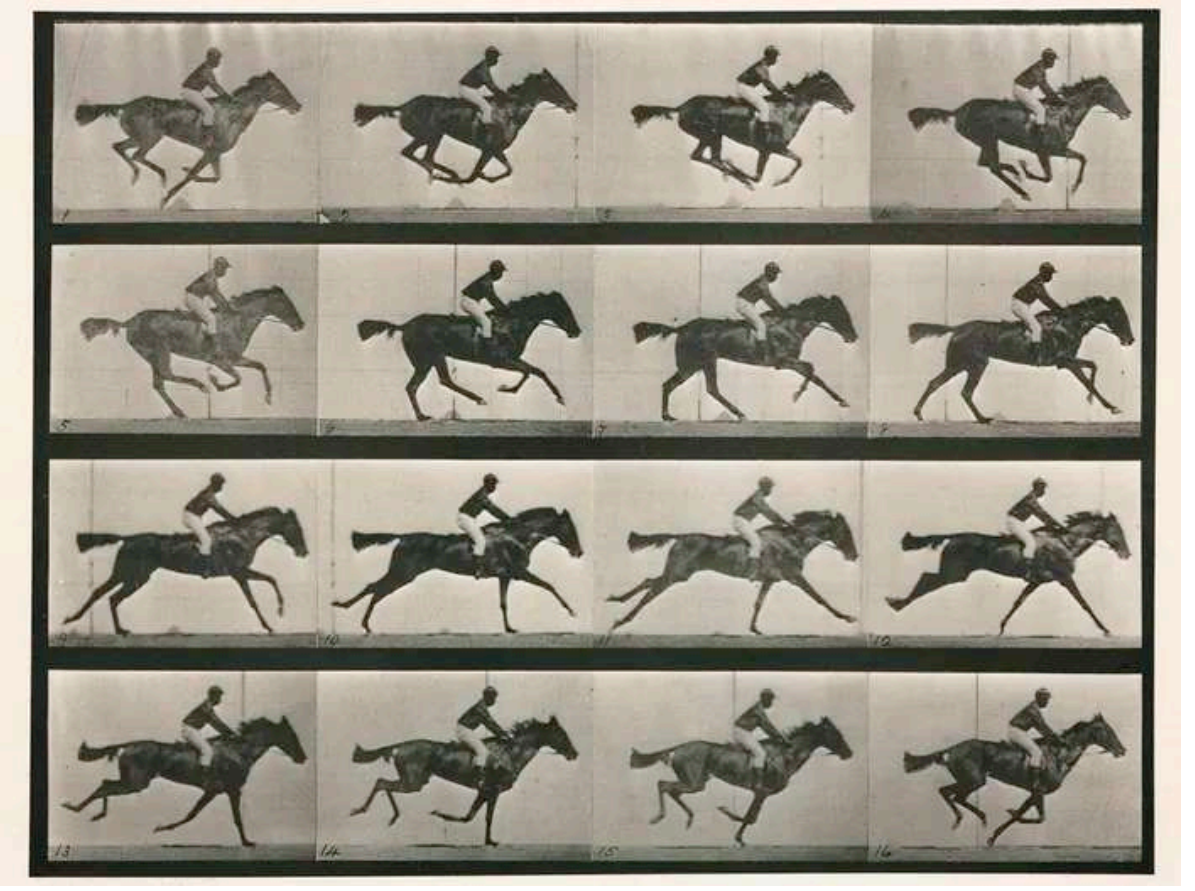


Dynamic Perspective

- Motion. Video.

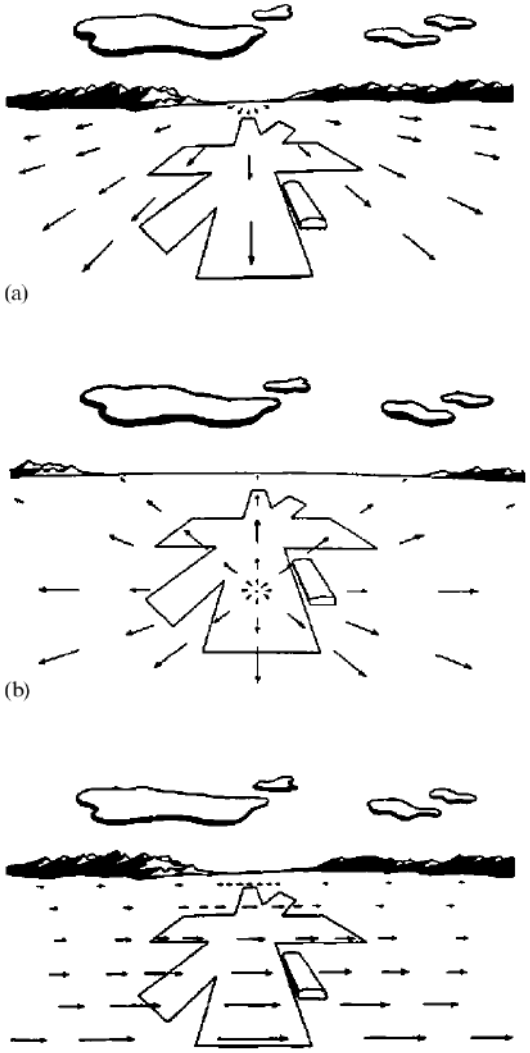
Eadweard Muybridge





Dynamic Perspective

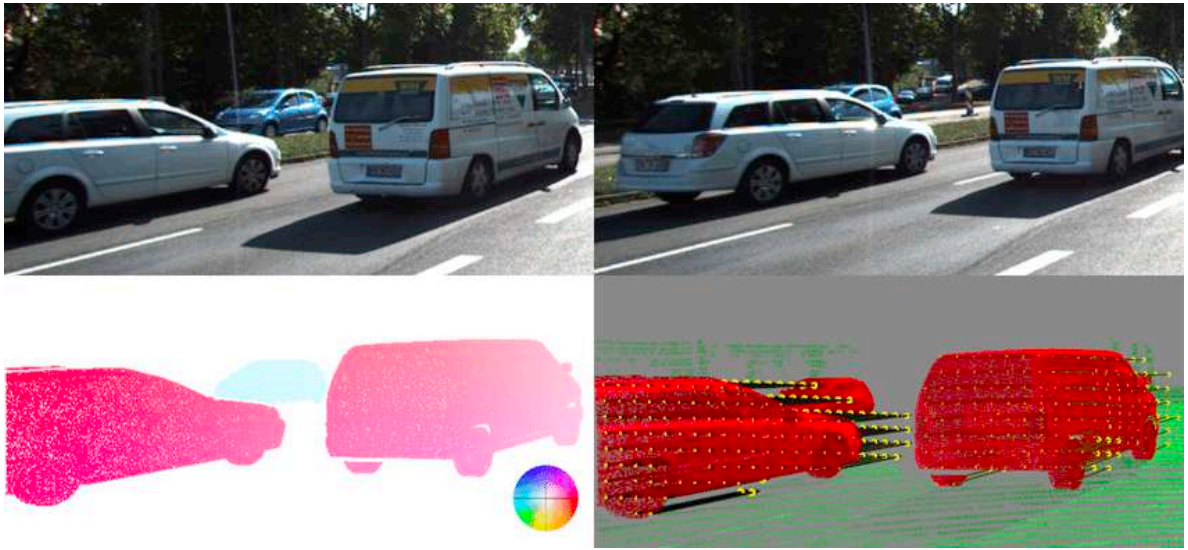
- Optical Flow.
- J J Gibson's examples:



Zamir

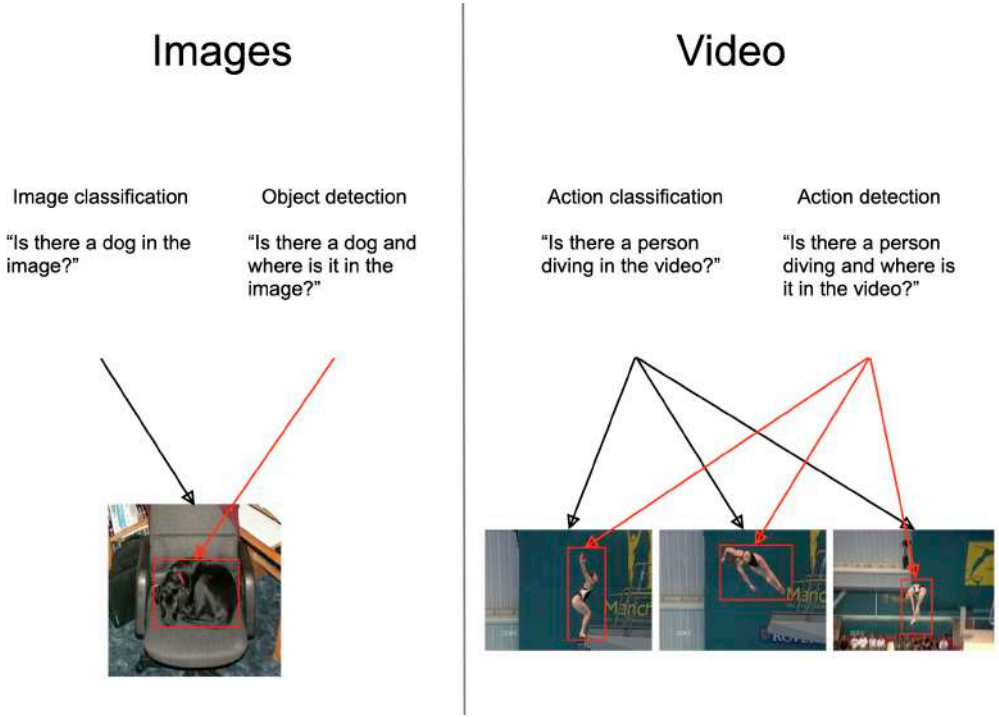
Dynamic Perspective

- Optical Flow



Dynamic Perspective

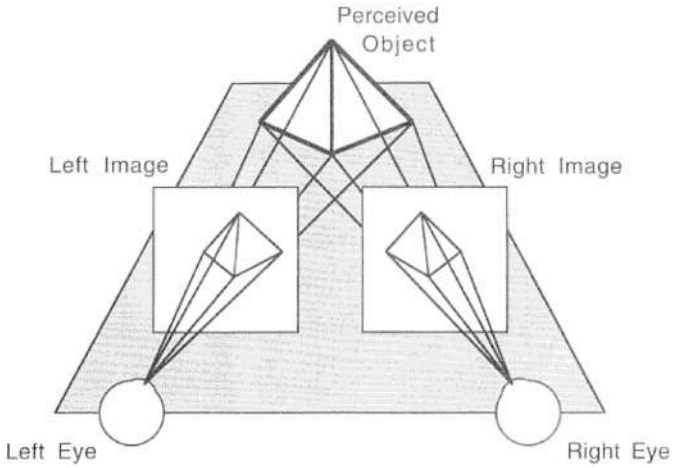
- Recognition in Videos
  - E.g. actions
    - Though actions are not exclusive to videos.



Dynamic Perspective→3D

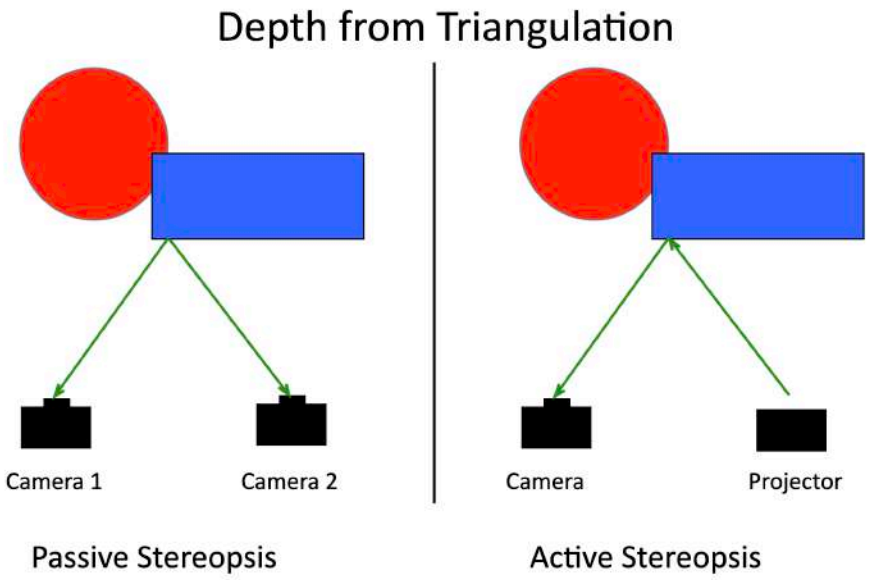
- Moving (or multi) Camera

Binocular Stereopsis

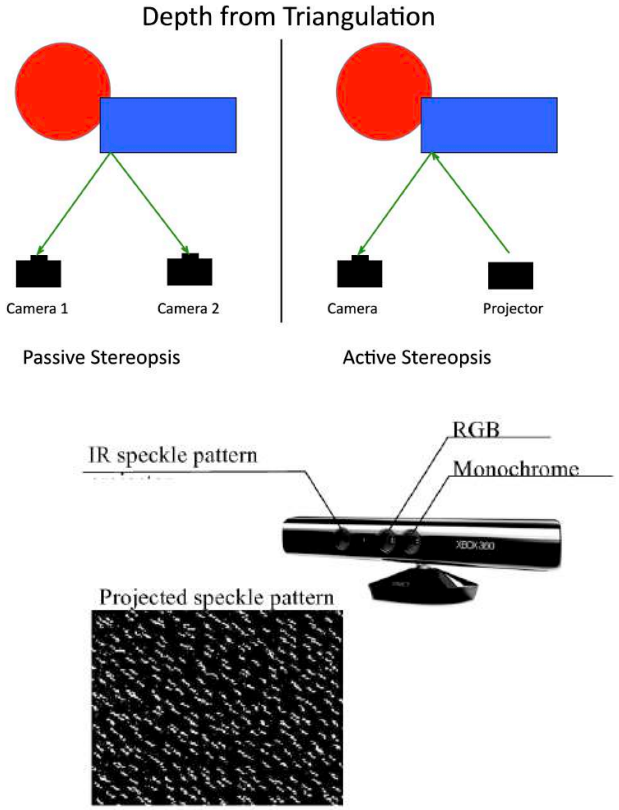


3D

- Moving (or multi) Camera

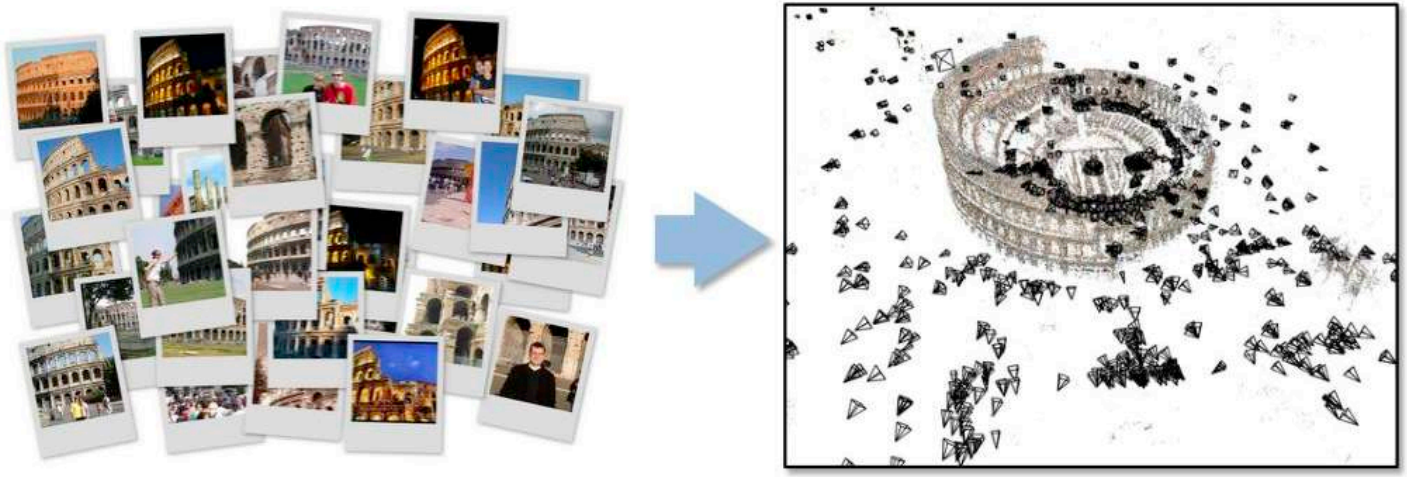


3D


Depth from Triangulation
Camera 1
Camera 2
Camera
Projector
Passive Stereopsis
Active Stereopsis
IR speckle pattern
RGB
Monochrome
XBOX360
Projected speckle pattern

3D

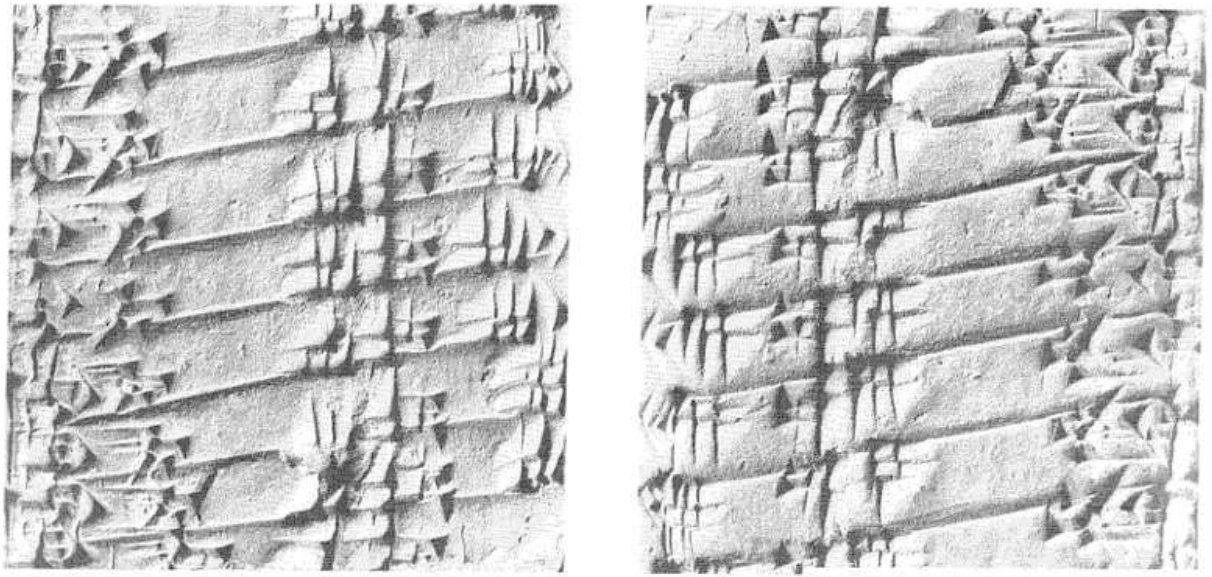
- Moving (or multi) Camera



Building Rome in a Day. Snavely et al.

Other 3D estimation cues

- e.g. Pictorial cues (shading)



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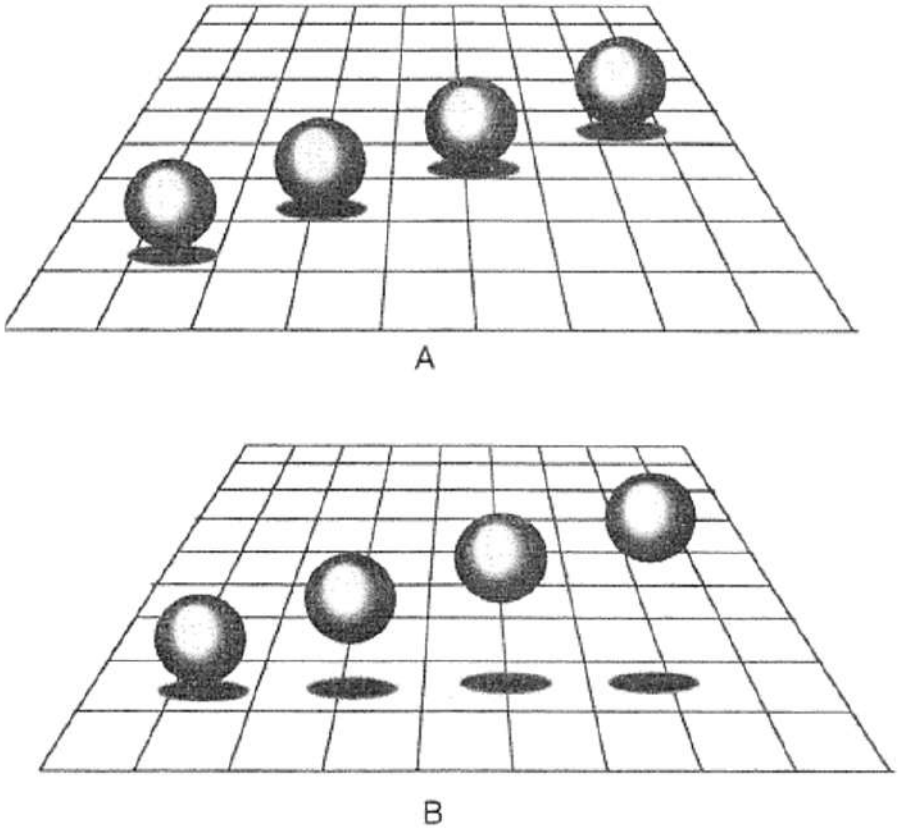
Other 3D estimation cues

- e.g. Pictorial cues (shadows)



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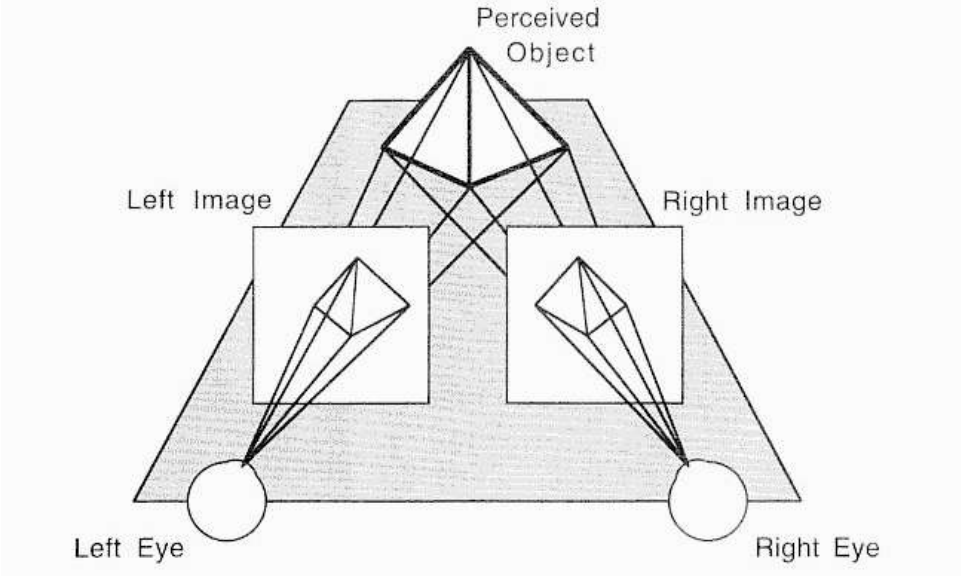
List of 3D cues (Vision Science, Palmer)

Zamir

INFORMATION SOURCE	Ocular/ Optical	Binocular/ Monocular	Static/ Dynamic	Relative/ Absolute	Qualitative/ Quantitative
Accommodation	ocular	monocular	static	absolute	quantitative
Convergence	ocular	binocular	static	absolute	quantitative
Binocular Disparity	optical	binocular	static	relative	quantitative
Motion Parallax	optical	monocular	dynamic	relative	quantitative
Texture Accretion/Deletion	optical	monocular	dyanmic	relative	qualitative
Convergence of Parallels	optical	monocular	static	relative	quantitative
Position relative to Horizon	optical	monocular	static	relative	quantitative
Relative Size	optical	monocular	static	relative	quantitative
Familiar Size	optical	monocular	static	absolute	quantitative
Texture Gradients	optical	monocular	static	relative	quantitative
Edge Interpretation	optical	monocular	static	relative	qualitative
Shading and Shadows	optical	monocular	static	relative	qualitative
Aerial Perspective	optical	monocular	static	relative	qualitative

J. Malik

Binocular Stereopsis



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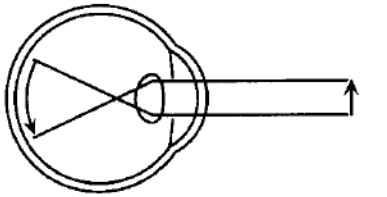
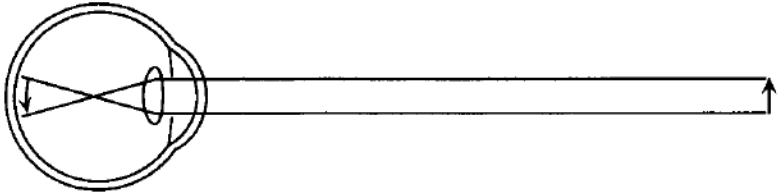
Accommodation



Thick Lens → Close



Thin Lens → Far

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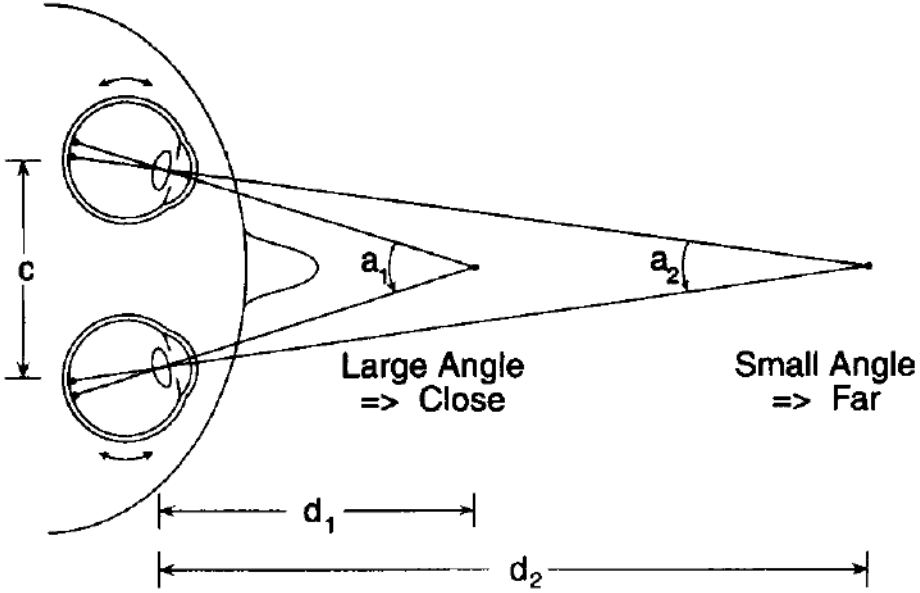
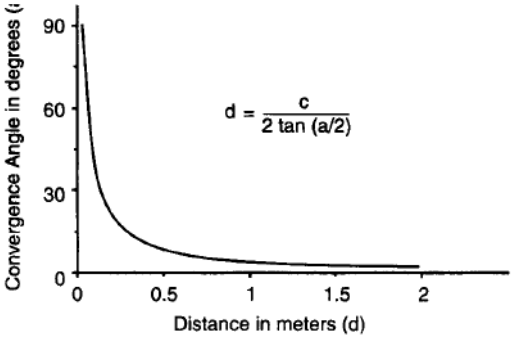
Convergence





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Texture gradient

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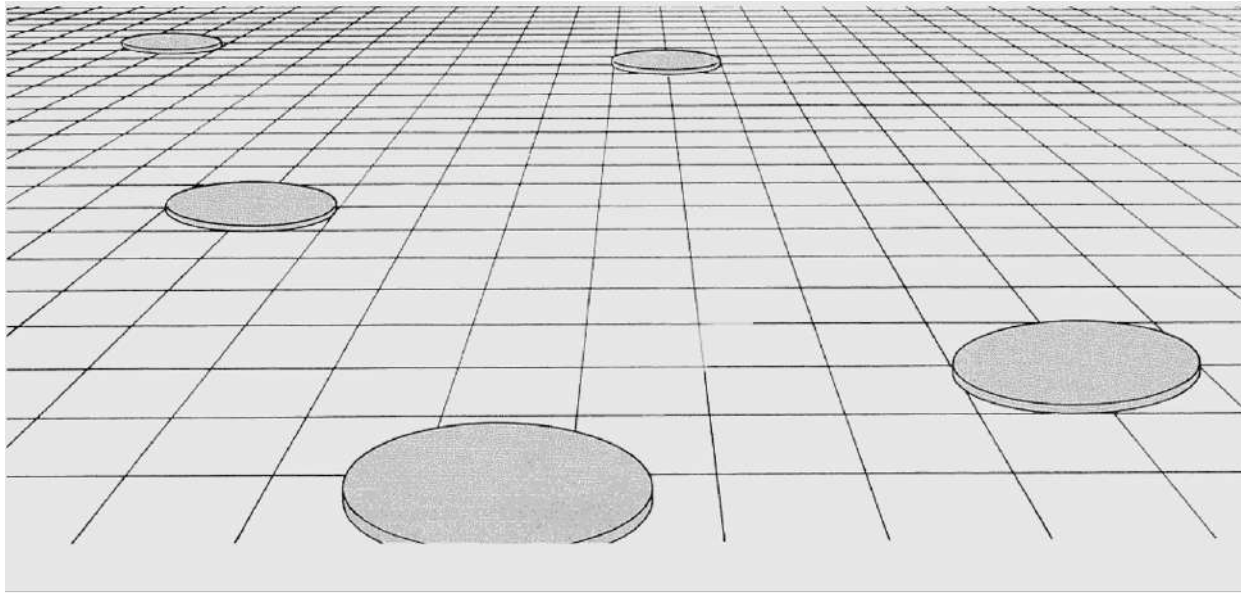
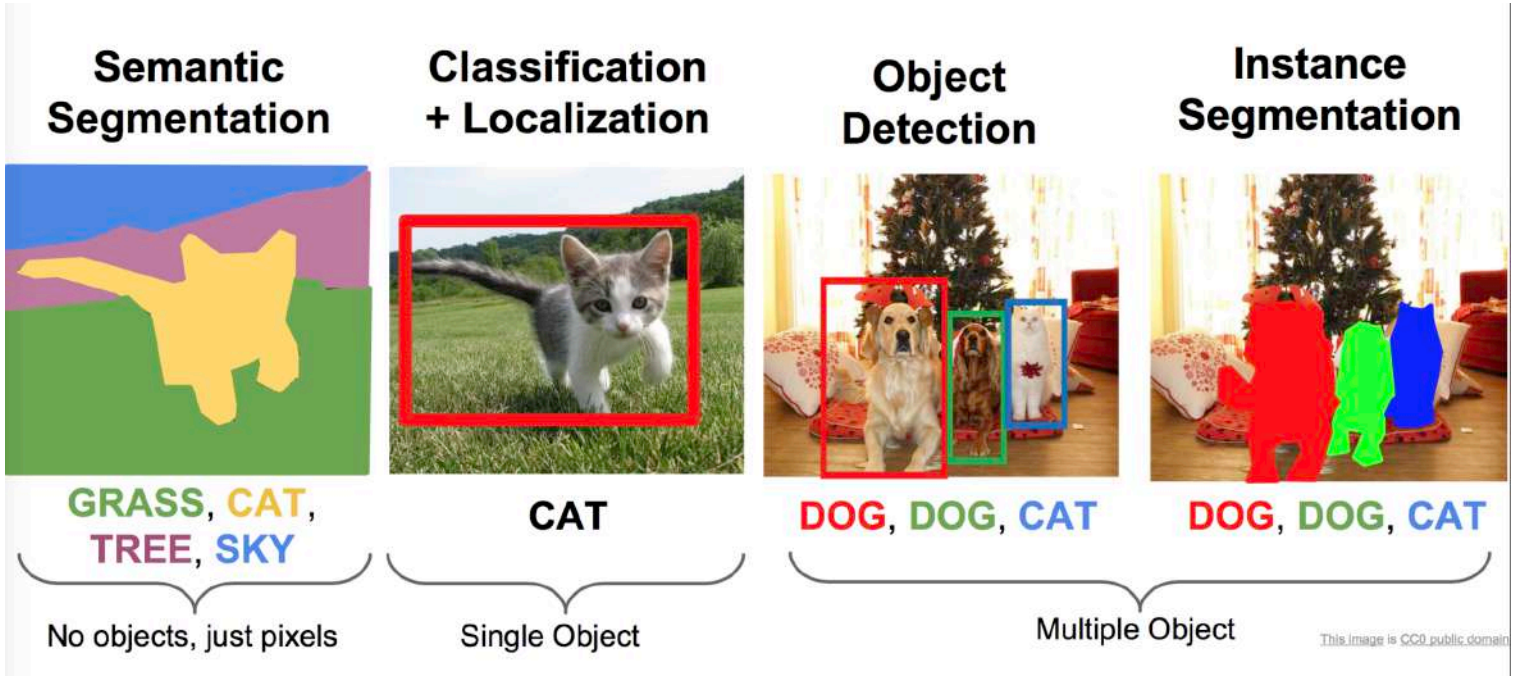
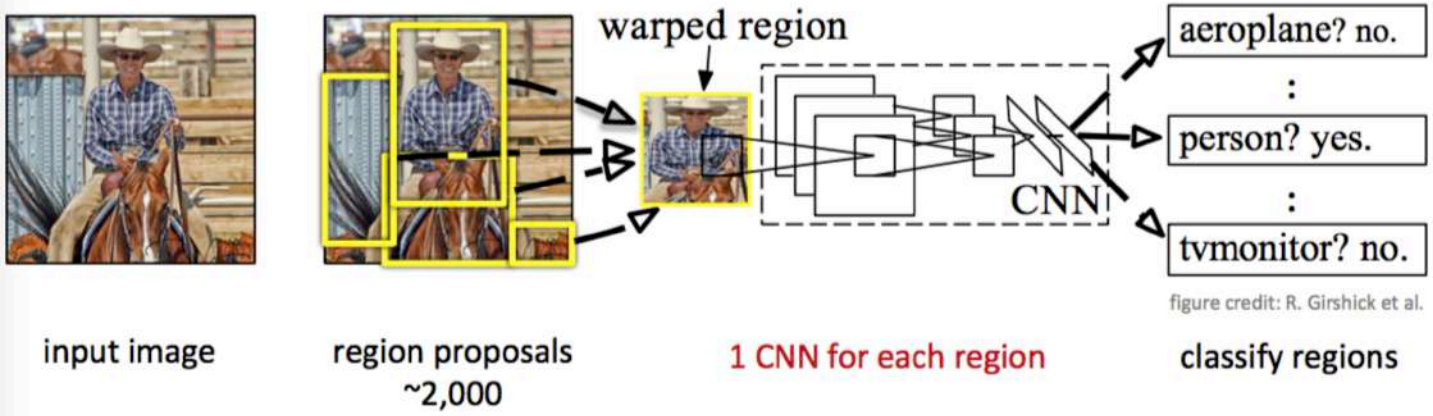


J. Malik

Recognition



Recognition

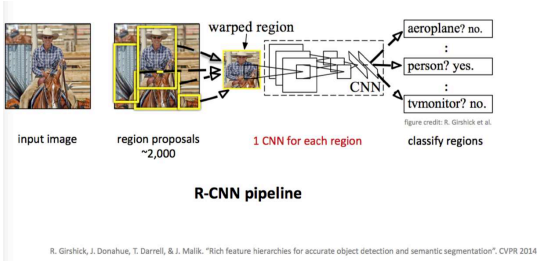


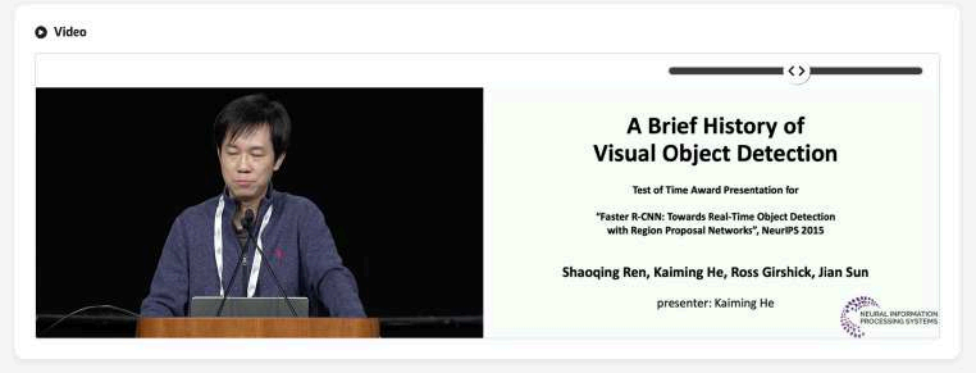
R-CNN pipeline

R. Girshick, J. Donahue, T. Darrell, & J. Malik. "Rich feature hierarchies for accurate object detection and semantic segmentation". CVPR 2014

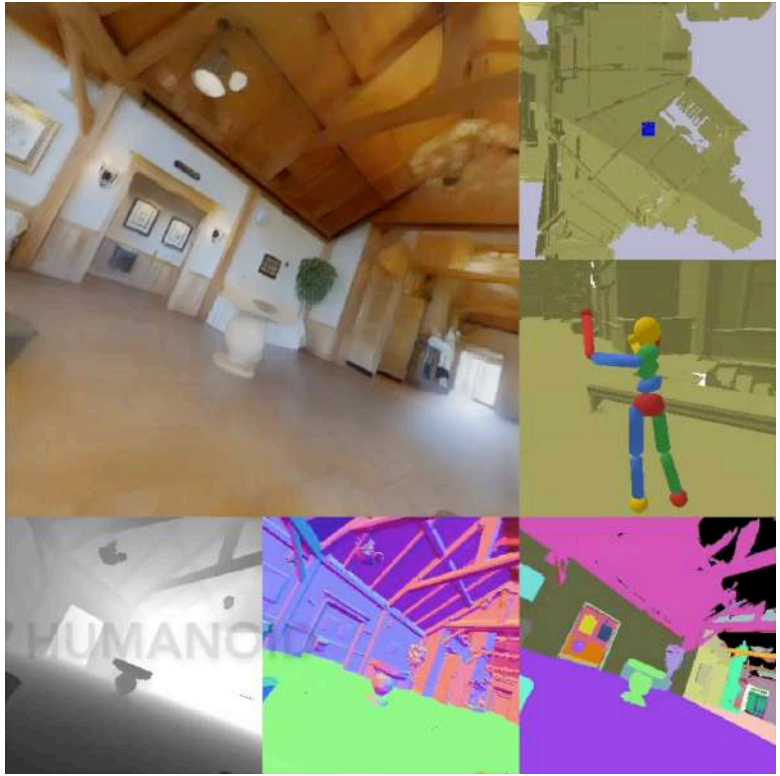
Recognition

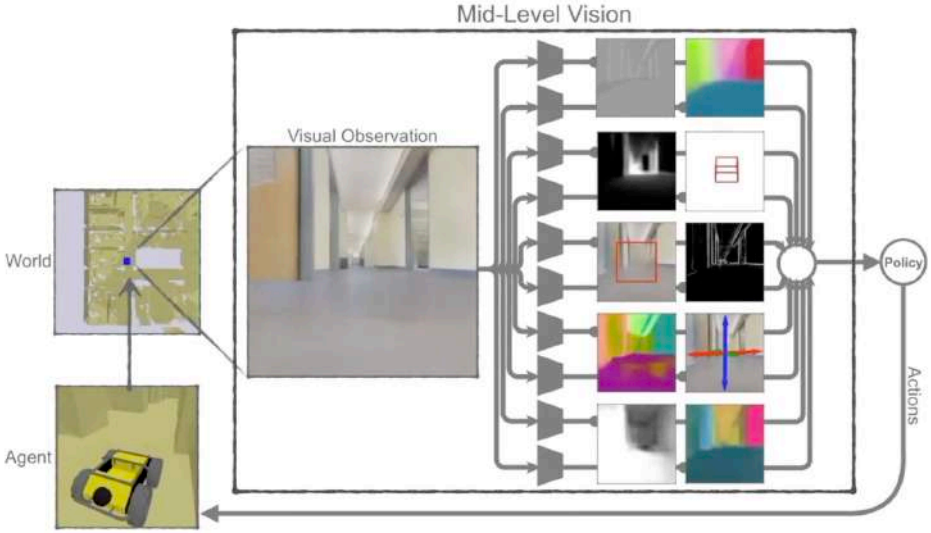
- NeuriPS’25 Test of Time Award talk: A Brief History of Visual Object Detection, Kaiming He
  - https://neurips.cc/virtual/2025/loc/san-diego/test-of-time/128328





Embodied/Active Vision





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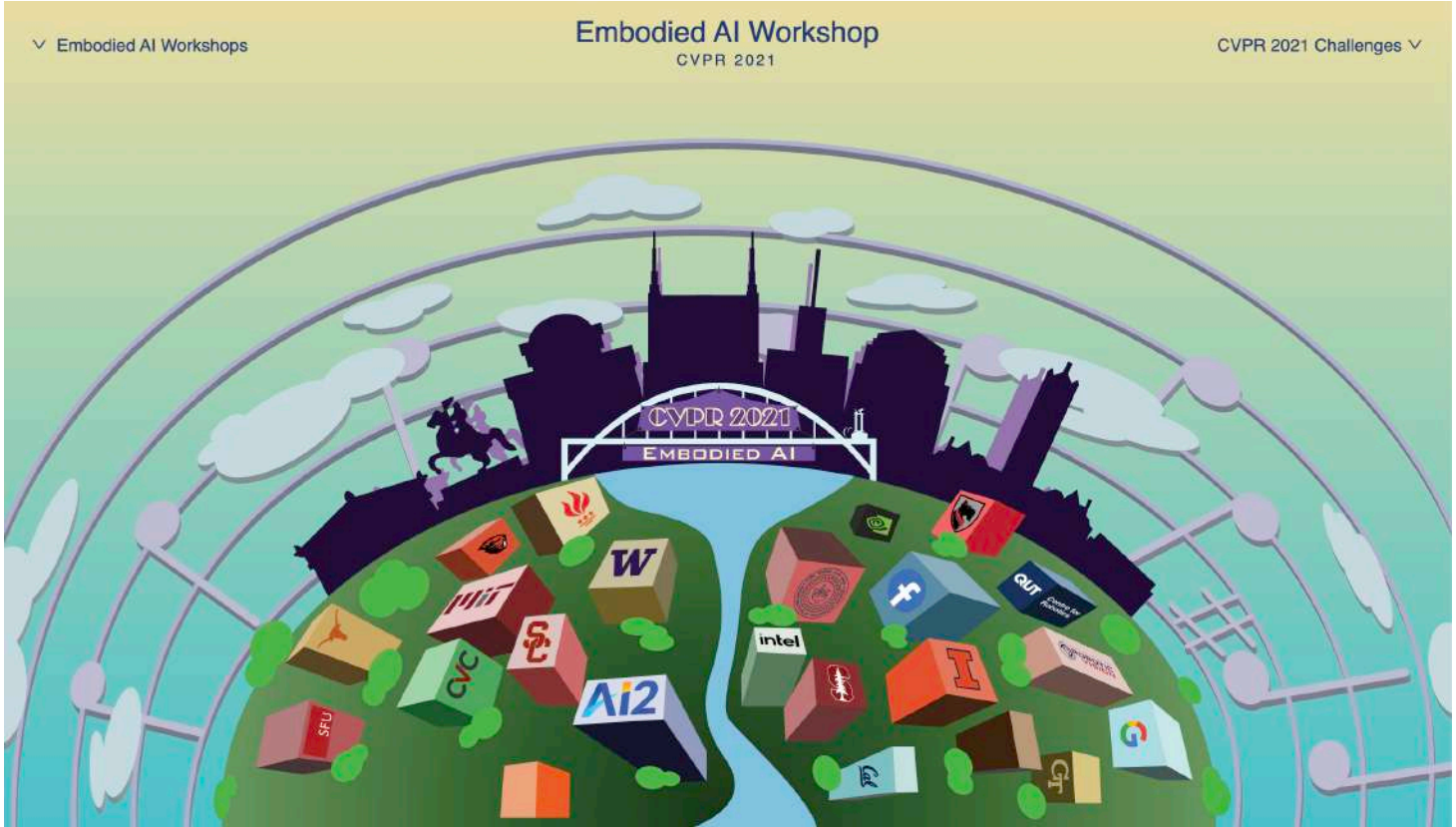
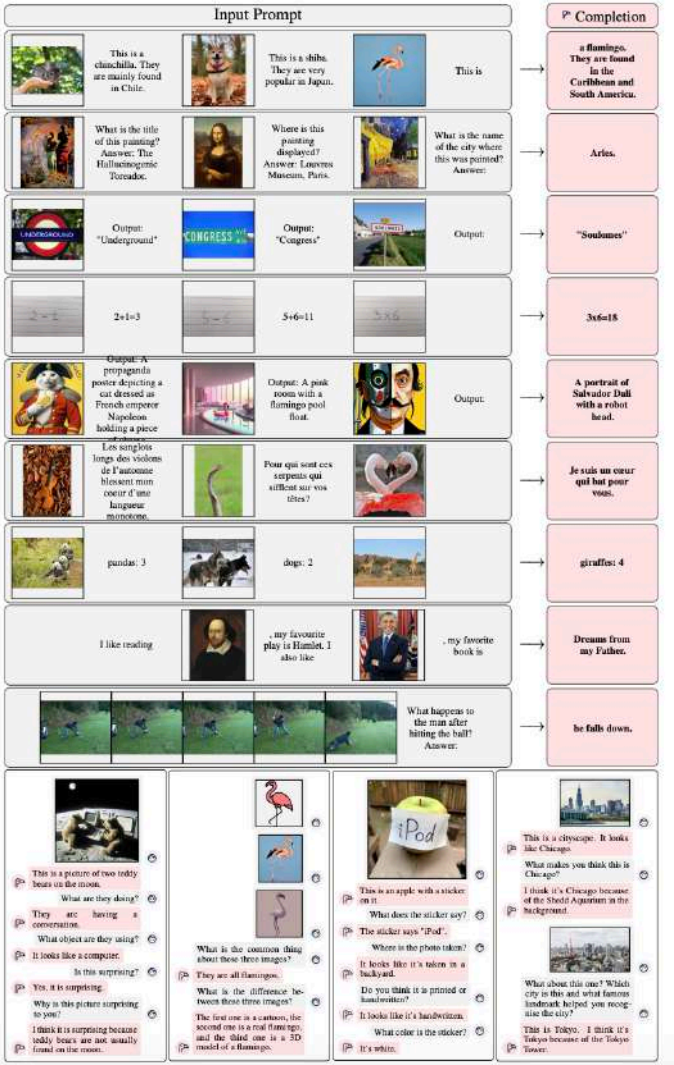
Embodied/Active Vision



“Foundation” models
Vision-Language Models
Multimodal Models



Flamingo,, Alayrac et al, 2022.
LLaVA, Liu et al, 2023
4M, Mizrahi & Bachmann et al. 2023.

“Foundation” models
Vision-Language Models
Multimodal Models

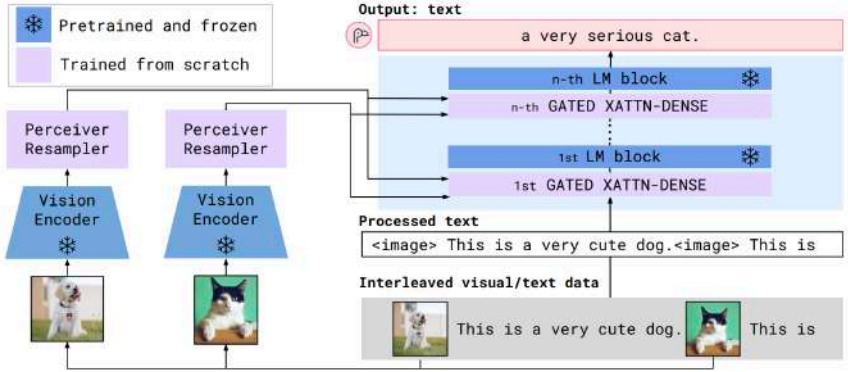
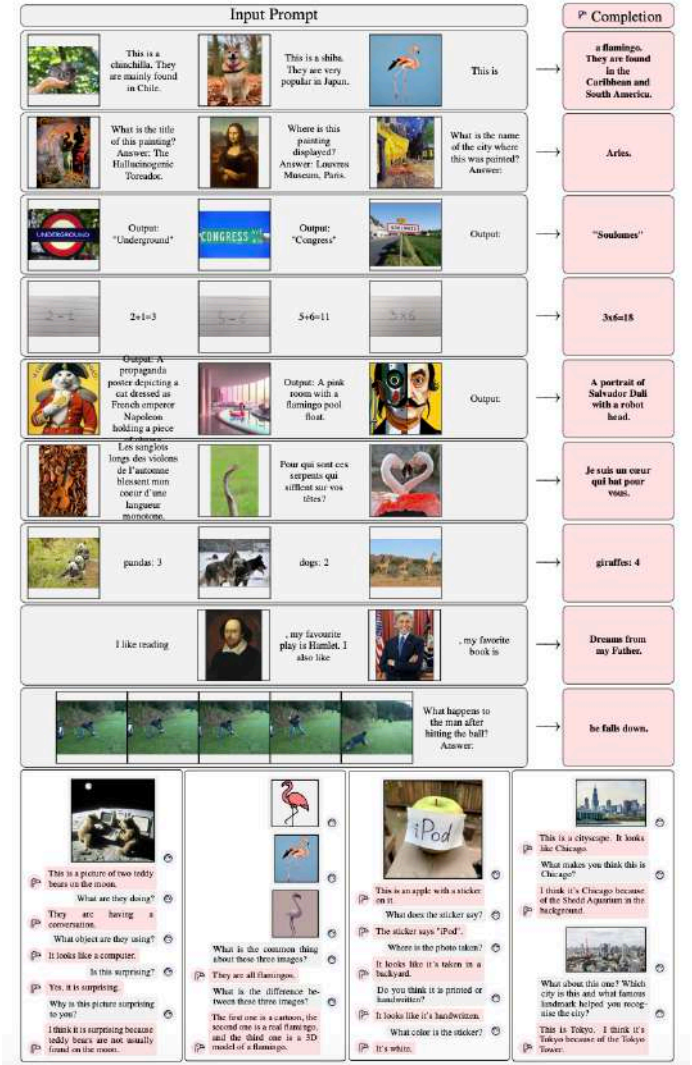
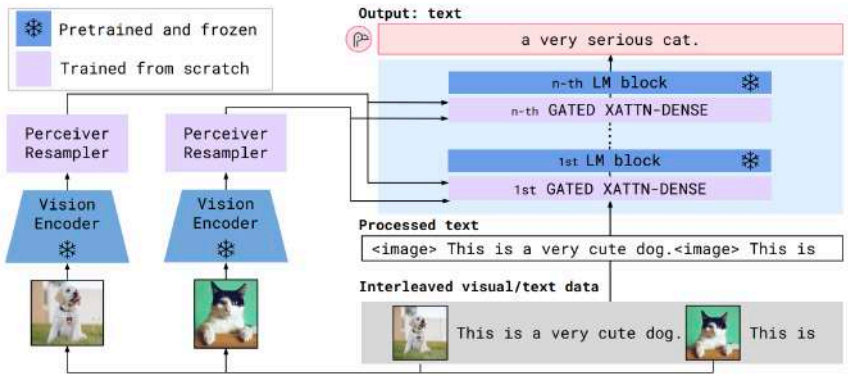


Figure 3: **Flamingo architecture overview.** Flamingo is a family of visual language models (VLMs) that take as input visual data interleaved with text and produce free-form text as output.



Flamingo,, Alayrac et al, 2022.
LLaVA, Liu et al, 2023
4M, Mizrahi & Bachmann et al. 2023.

“Foundation” models
Vision-Language Models
Multimodal Models



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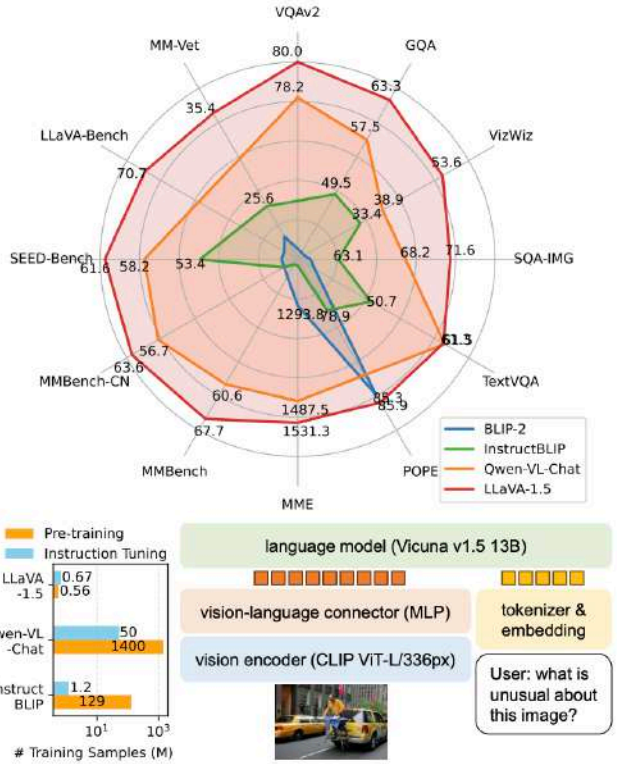

Flamingo,, Alayrac et al, 2022.
LLaVA, Liu et al, 2023
4M, Mizrahi & Bachmann et al. 2023.

“Foundation” models
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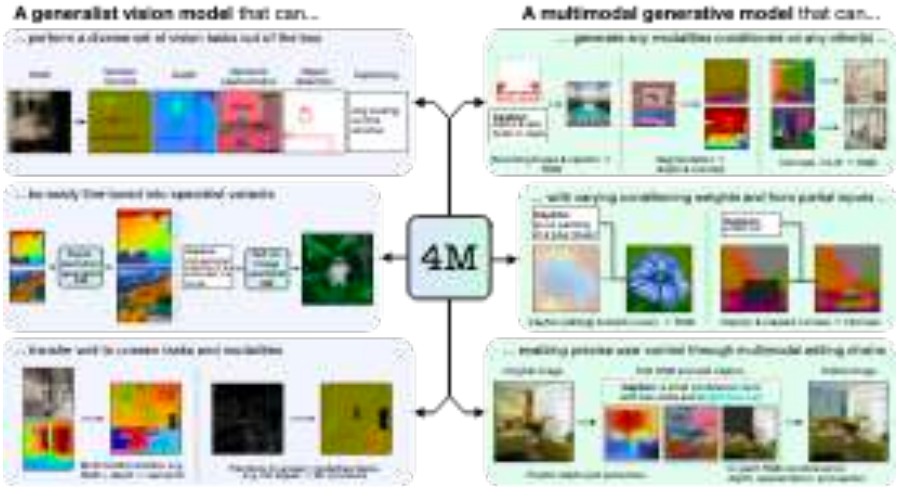
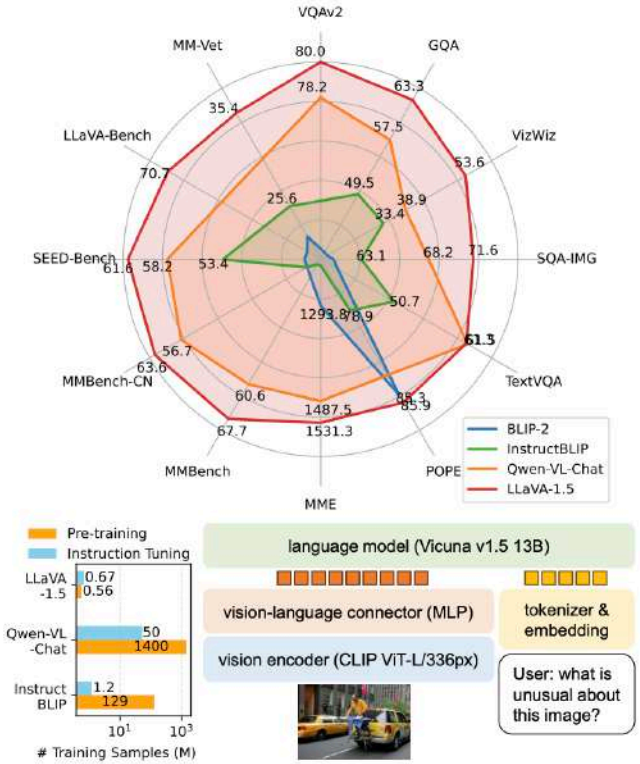




Flamingo,, Alayrac et al, 2022.
LLaVA, Liu et al, 2023
4M, Mizrahi & Bachmann et al. 2023.

Image Processing Perspective

- e.g. White Balancing

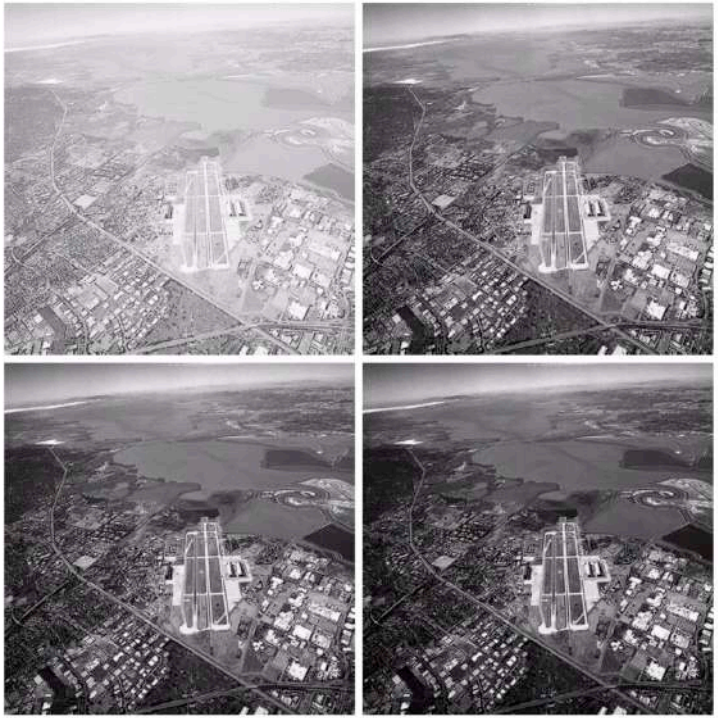
- Manual
  - Choose color-neutral object in the photos and normalize
- Automatic (AWB)
  - Grey World: force average color of scene to grey
  - White World: force brightest object to white

Image Processing Perspective

- Point Processing

a b
c d

FIGURE 3.9 (a) Aerial image. (b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 3.0, 4.0$, and 5.0, respectively. (Original image for this example courtesy of NASA.)



The simplest kind of range transformations are these independent of position x,y:

$$g = T(f)$$

This is called point processing.

e.g. Gain and Bias transform:

$$g(x,y) = a*f(x,y) + b$$

A. Efros.

Image Processing Perspective

Original

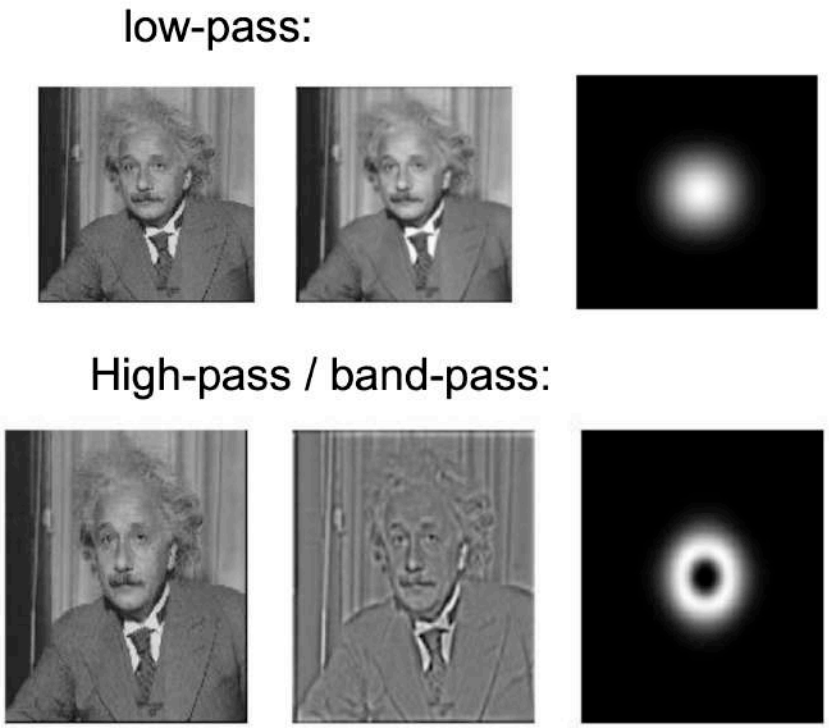
$$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} =$$

Blur (with a mean filter)

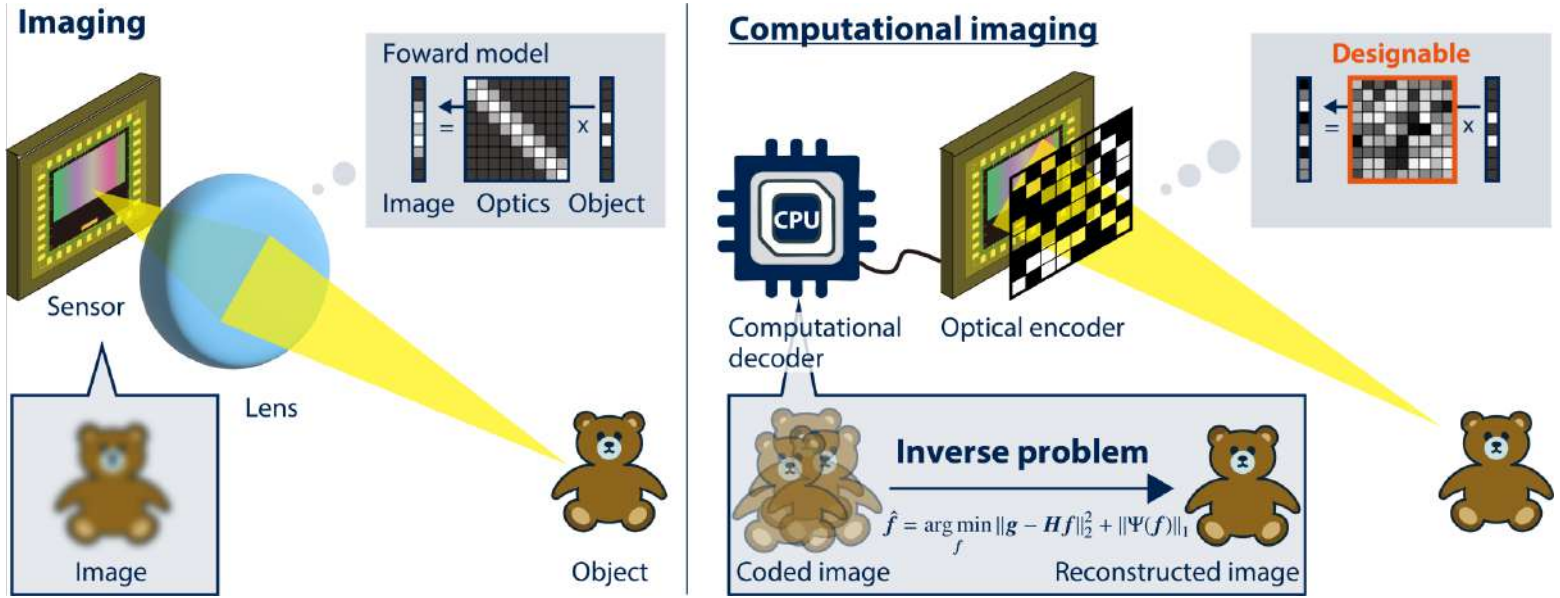
A. Efros.

Image Processing Perspective

- e.g. Frequency Analyses

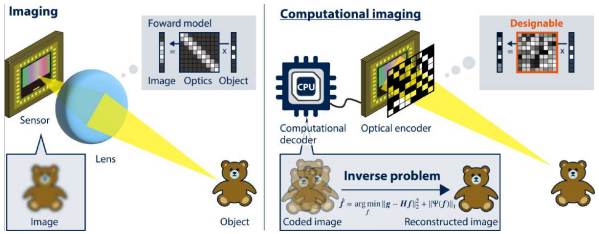


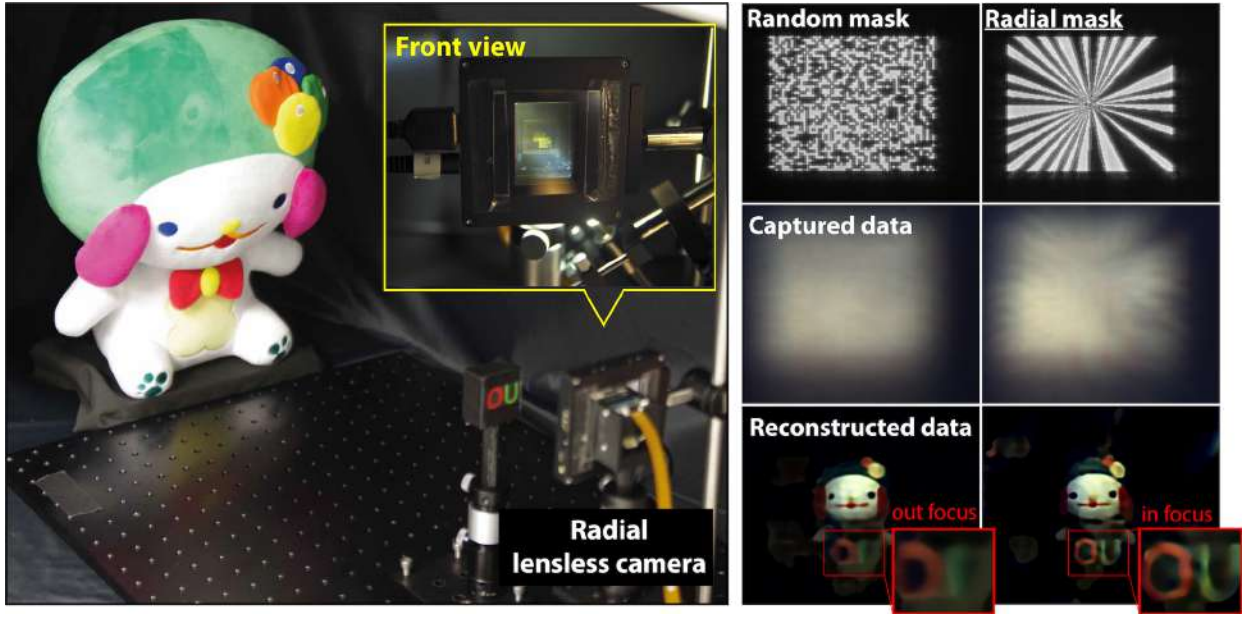
Computational Photography



http://www.am.sanken.osaka-u.ac.jp/research_/ci-en/

Computational Photography





http://www.am.sanken.osaka-u.ac.jp/research_/ci-en/

Not in the recap (In SOTA lectures)

- Neural Network architecture
- Embodied vision
- Foundation models: language, multimodal, generative, etc. models.
- …

Some left out elephants

- Plenoptic function -> Radiance Fields -> Neural Radiance Fields

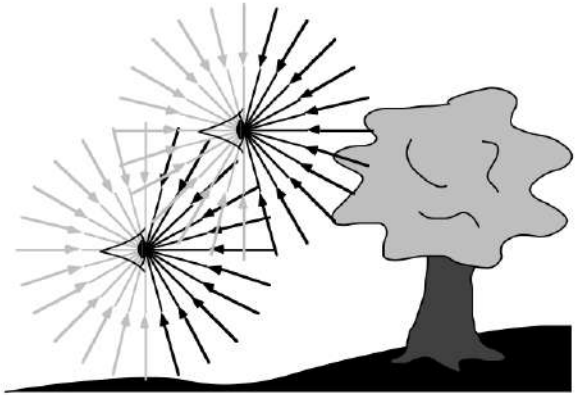
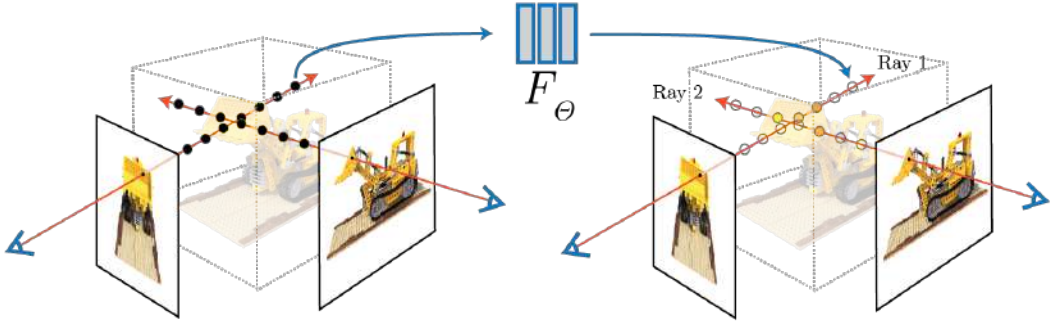


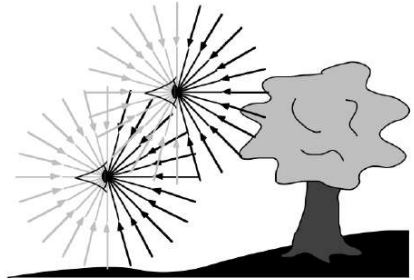
Fig.1.3
The plenoptic function describes the information available to an observer at any point in space and time. Shown here are two schematic eyes-which one should consider to have punctate pupils-gathering pencils of light rays. A real observer cannot see the light rays coming from behind, but the plenoptic function does include these rays.

Zamir



Some left out elephants

- Plenoptic function -> Radiance Fields -> Neural Radiance Fields
- 3D Gaussain Splatting



Zamir

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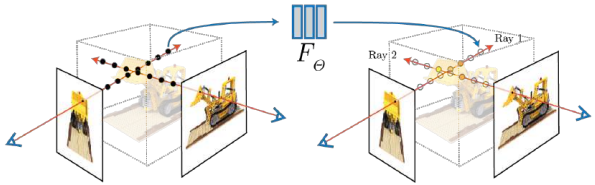
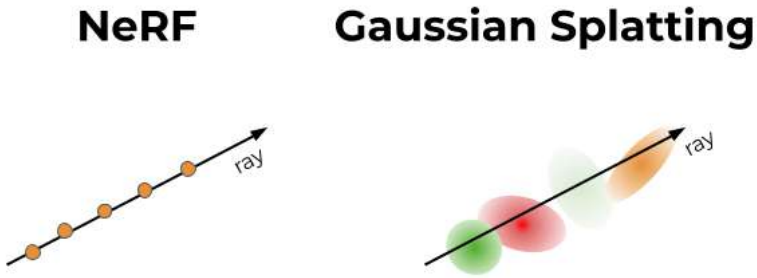
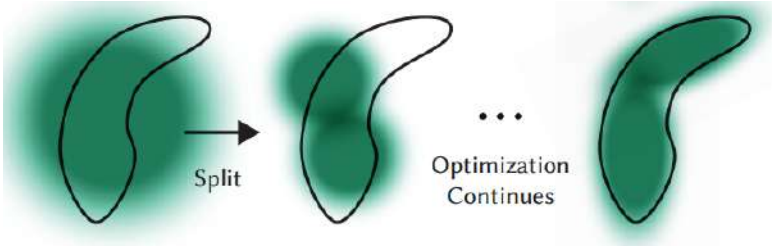
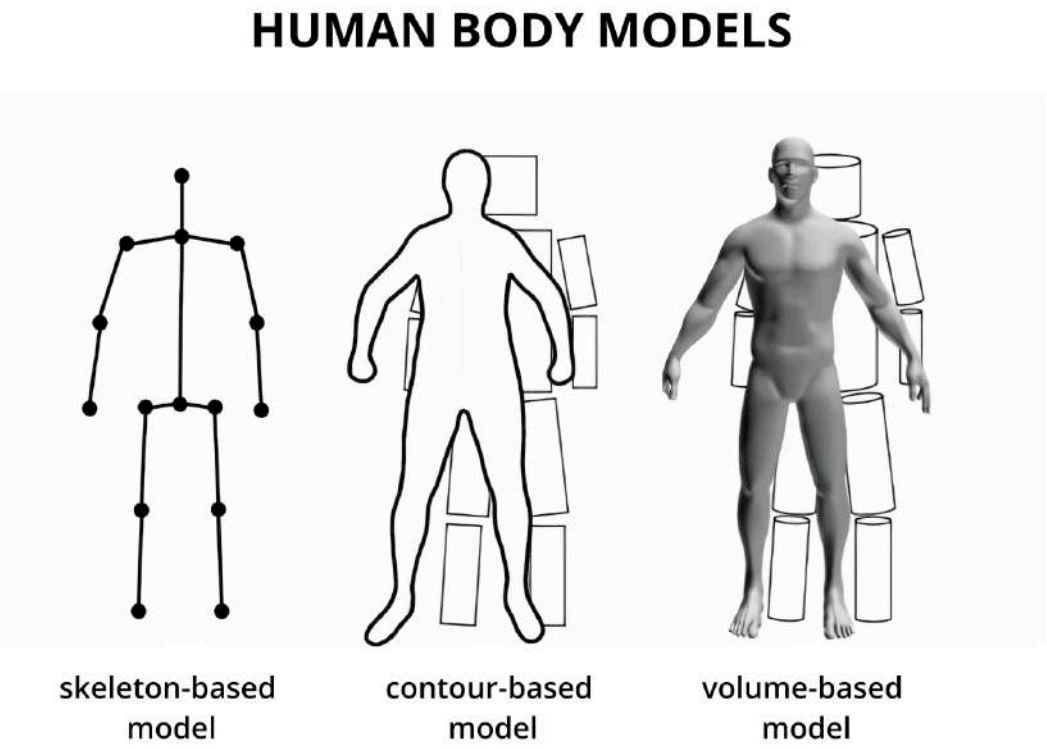


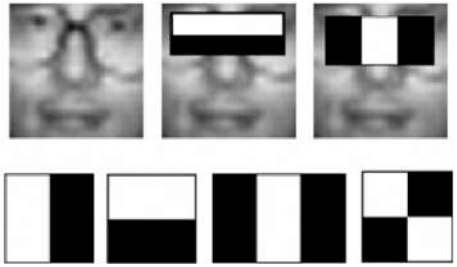




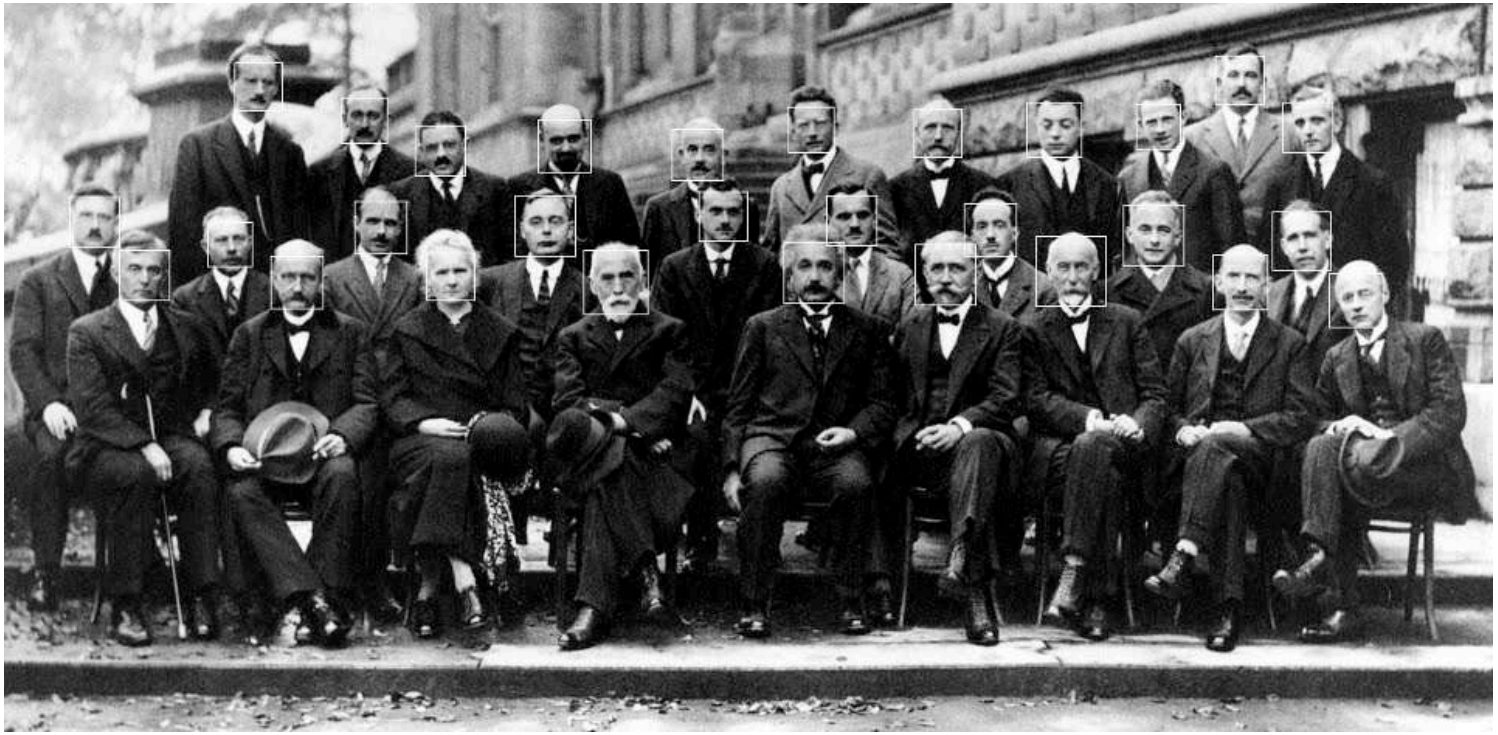
other (more specific) problems



other (more specific) problems

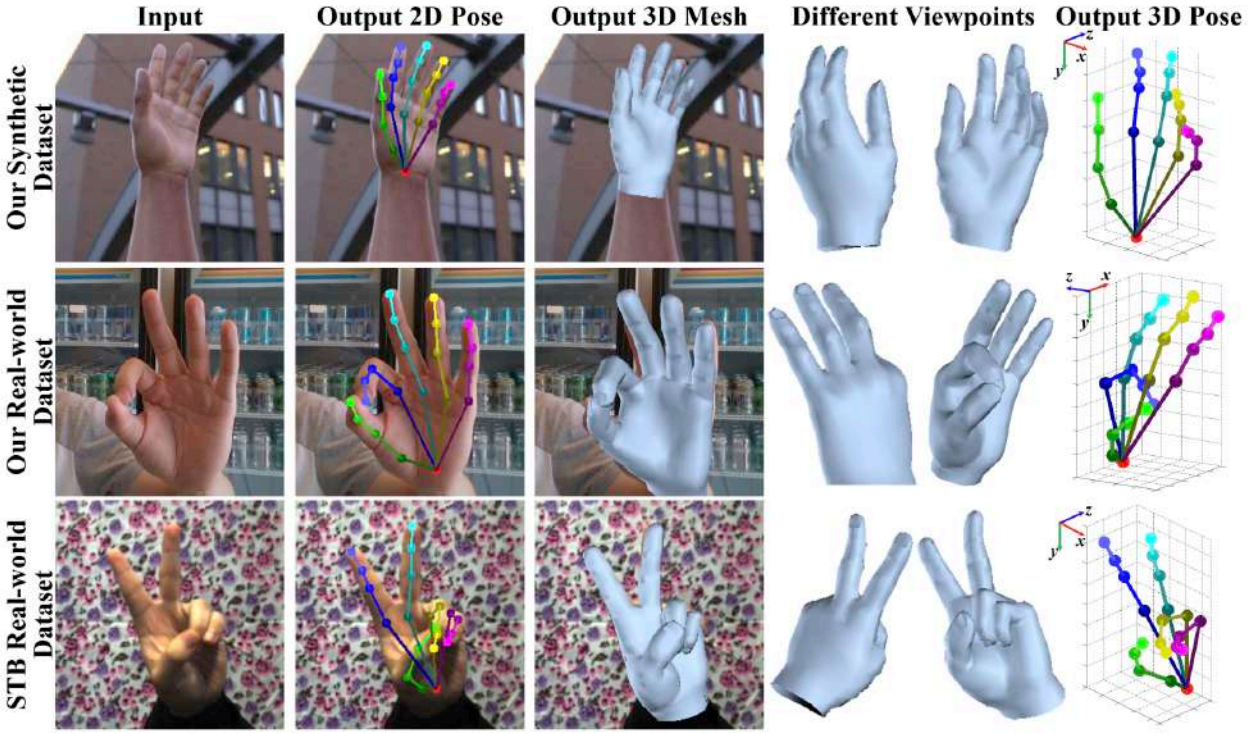


Viola & Jones



Zamir

other (more specific) problems



Ge et al, 2019

Questions?

https://vilab.epfl.ch/